At the roots of lean and agility

Essays on Lean and Agile Software Engineering

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Prologue

Every chapter that follows opens with a comic strip narrated by Captain Lucian, a character built on the real Lucian of Samosata, who opened his own *A True Story* around 170 CE by explaining, in his own words, exactly why he was about to lie to his readers.

Original text [37]	**English translation [38]**
Ὥσπερ τοῖς ἀθλητικοῖς καὶ περὶ τὴν τῶν σωμάτων ἐπιμέλειαν ἀσχολουμένοις οὐ τῆς εὐεξίας μόνον οὐδὲ τῶν γυμνασίων φροντίς ἐστιν, ἀλλὰ καὶ τῆς κατά καιρὸν γινομένης ἀνέσεως — μέρος γοῦν τῆς ἀσκήσεως τὸ μέγιστον αὐτὴν ὑπολαμβάνουσιν — οὕτω δὴ καὶ τοῖς περὶ τοὺς λόγους ἐσπουδακόσιν ἡγοῦμαι προσήκειν μετὰ τὴν πολλὴν τῶν σπουδαιοτέρων ἀνάγνωσιν ἀνιέναι τε τὴν διάνοιαν καὶ πρὸς τὸν ἔπειτα κάματον ἀκμαιοτέραν παρασκευάζειν.	Just as athletes and those who train their bodies take care to make themselves strong not only through exertion but also, from time to time, through rest, which is considered a very important part of physical training, so too, I think, those devoted to study should rest their minds after serious reading, so that they may return to their work refreshed.
γένοιτο δ᾽ ἂν ἐμμελὴς ἡ ἀνάπαυσις αὐτοῖς, εἰ τοῖς τοιούτοις τῶν ἀναγνωσμάτων ὁμιλοῖεν, ἃ μὴ μόνον ἐκ τοῦ ἀστείου τε καὶ χαρίεντος ψιλὴν παρέξει τὴν ψυχαγωγίαν, ἀλλά τινα καὶ θεωρίαν οὐκ ἄμουσον ἐπιδείξεται, οἷόν τι καὶ περὶ τῶνδε τῶν συγγραμμάτων φρονήσειν ὑπολαμβάνω· οὐ γὰρ μόνον τὸ ξένον τῆς ὑποθέσεως οὐδὲ τὸ χαρίεν τῆς προαιρέσεως ἐπαγωγὸν ἔσται αὐτοῖς οὐδ᾽ ὅτι ψεύσματα ποικίλα πιθανῶς τε καὶ ἐναλήθως ἐξενηνόχαμεν, ἀλλ᾽ ὅτι καὶ τῶν ἱστορουμένων ἕκαστον οὐκ ἀκωμῳδήτως ἥνικται πρός τινας τῶν παλαιῶν ποιητῶν τε καὶ συγγραφέων καὶ φιλοσόφων πολλὰ τεράστια καὶ μυθώδη συγγεγραφότων.	They will find suitable relaxation in reading works that are enjoyable both for their charm and wit and for the lessons they contain, presented with a certain elegance. I hope that this work of mine will be regarded in just this way. It should be enjoyable not only because of the unusual structure of its subject and the liveliness of its ideas, and because I have filled it with many inventions that seem plausible and convincing, but also because every absurdity I relate is a comic allusion to certain ancient poets, historians, and philosophers who wrote so many fabulous and astonishing stories.

Original text [37]	**English translation [38]**
Κτησίας ὁ Κτησιόχου ὁ Κνίδιος, ὃς συνέγραψεν περὶ τῆς Ἰνδῶν χώρας καὶ τῶν παρ᾽ αὐτοῖς ἃ μήτε αὐτὸς εἶδεν μήτε ἄλλου ἀληθεύοντος ἤκουσεν. ἔγραψε δὲ καὶ Ἰαμβοῦλος περὶ τῶν ἐν τῇ μεγάλῃ θαλάττῃ πολλὰ παράδοξα, γνώριμον μὲν ἅπασι τὸ ψεῦδος πλασάμενος, οὐκ ἀτερπῆ δὲ ὅμως συνθεὶς τὴν ὑπόθεσιν. πολλοὶ δὲ καὶ ἄλλοι τὰ αὐτὰ τούτοις προελόμενοι συνέγραψαν ὡς δή τινας ἑαυτῶν πλάνας τε καὶ ἀποδημίας, θηρίων τε μεγέθη ἱστοροῦντες καὶ ἀνθρώπων ὠμότητας καὶ βίων καινότητας· ἀρχηγὸς δὲ αὐτοῖς καὶ διδάσκαλος τῆς τοιαύτης βωμολοχίας ὁ τοῦ Ὁμήρου Ὀδυσσεύς, τοῖς περὶ τὸν Ἀλκίνουν διηγούμενος ἀνέμων τε δουλείαν καὶ μονοφθάλμους καὶ ὠμοφάγους καὶ ἀγρίους τινὰς ἀνθρώπους, ἔτι δὲ πολυκέφαλα ζῷα καὶ τὰς ὑπὸ φαρμάκων τῶν ἑταίρων μεταβολάς, οἷα πολλὰ ἐκεῖνος πρὸς ἰδιώτας ἀνθρώπους τοὺς Φαίακας ἐτερατεύσατο.	Ctesias, son of Ctesiochus of Cnidus, wrote about India things that he had neither seen himself nor heard from anyone else. Iambulus wrote many extraordinary stories about what could find in the great ocean; and although he filled it up with lies that everyone could recognize as such, still, his work is not unpleasant to read. Many others did the same, writing about journeys and travels to distant lands that never took place, telling of enormous beasts, savage men, and strange customs. The leader and master of all such charlatanry was Homer's Odysseus, who at the court of Alcinous told of the winds held captive, of huge savage men with a single eye in the middle of their foreheads, of many-headed monsters, of companions transformed by magic, and of countless other lies that he poured out before those poor gullible Phaeacians.

Original text [37]	**English translation [38]**
τούτοις οὖν ἐντυχὼν ἅπασιν, τοῦ ψεύσασθαι μὲν οὐ σφόδρα τοὺς ἄνδρας ἐμεμψάμην, ὁρῶν ἤδη σύνηθες ὂν τοῦτο καὶ τοῖς φιλοσοφεῖν ὑπισχνουμένοις· ἐκεῖνο δὲ αὐτῶν ἐθαύμασα, εἰ ἐνόμιζον λήσειν οὐκ ἀληθῆ συγγράφοντες. διόπερ καὶ αὐτὸς ὑπὸ κενοδοξίας ἀπολιπεῖν τι σπουδάσας τοῖς μεθ᾽ ἡμᾶς, ἵνα μὴ μόνος ἄμοιρος ὦ τῆς ἐν τῷ μυθολογεῖν ἐλευθερίας, ἐπεὶ μηδὲν ἀληθὲς ἱστορεῖν εἶχον — οὐδὲν γὰρ ἐπεπόνθειν ἀξιόλογον — ἐπὶ τὸ ψεῦδος ἐτραπόμην πολὺ τῶν ἄλλων εὐγνωμονέστερον· κἂν ἓν γὰρ δὴ τοῦτο ἀληθεύσω λέγων ὅτι ψεύδομαι. οὕτω δ᾽ ἄν μοι δοκῶ καὶ τὴν παρὰ τῶν ἄλλων κατηγορίαν ἐκφυγεῖν αὐτὸς ὁμολογῶν μηδὲν ἀληθὲς λέγειν. γράφω τοίνυν περὶ ὧν μήτε εἶδον μήτε ἔπαθον μήτε παρ᾽ ἄλλων ἐπυθόμην, ἔτι δὲ μήτε ὅλως ὄντων μήτε τὴν ἀρχὴν γενέσθαι δυναμένων. διὸ δεῖ τοὺς ἐντυγχάνοντας μηδαμῶς πιστεύειν αὐτοῖς.	Whenever I came across writers of this kind, I did not blame them too severely for telling lies, since I saw that philosophers too have long been accustomed to doing the same. What surprised me was that they imagined they could make us swallow their stories as truth. So I too felt the urge to leave something behind for posterity, since I did not want to be the only person denied the freedom to tell tall tales. And since I have nothing true to relate — because nothing has ever happened to me that deserves to be told — I have turned to fiction. My lie, however, is much more reasonable than theirs, because at least I will tell one truth: I am going to lie. In this way, perhaps I shall escape the criticism deserved by the others, since I openly admit that I am not telling the truth at all. I therefore write about things that I have never experienced or even seen, that I have never learned from anyone else, that do not exist, and that could never exist. Readers, therefore, should not believe a word of them.
Ὁρμηθεὶς γάρ ποτε ἀπὸ Ἡρακλείων στηλῶν καὶ ἀφεὶς εἰς τὸν ἑσπέριον ὠκεανὸν οὐρίῳ ἀνέμῳ τὸν πλοῦν ἐποιούμην. αἰτία δέ μοι τῆς ἀποδημίας καὶ ὑπόθεσις ἡ τῆς διανοίας περιεργία καὶ πραγμάτων καινῶν ἐπιθυμία καὶ τὸ βούλεσθαι μαθεῖν τί τὸ τέλος ἐστὶν τοῦ ὠκεανοῦ καὶ τίνες οἱ πέραν κατοικοῦντες ἄνθρωποι.	Once I have left the Pillars of Hercules and entered the western ocean, I continued on my voyage with a favorable wind. The curiosity moved me there with my desire to see new things, to discover where the ocean ends and what kind of people live on those distant places.

Quote 1: **A True Story**, Proem



Captain Lucian never files a course. Today he's provisioning for nowhere in particular – again.



Captain Lucian: "A merchant in the market swears he's seen the edge of the world. I intend to find out if he's lying."



Captain Lucian: "Fair warning – half of what we find, I won't understand myself until years later."



So they sail, chasing a rumor no sensible captain would trust.



His first mate has crossed this sea forty times, and trusts none of it.



His youngest hand has crossed it zero times, and trusts all of it.



Captain Lucian, tracing the chart: "Seven islands, if the merchant's map is honest. Each one stranger than the last."



None of them, it turns out, are quite what they first appear to be. Let's go find out why.

Figure 2: The chart

1
Introduction

In which the crew makes the decision to leave.

(Captain Lucian)



No cargo worth boasting about – just supplies for wherever the wind takes them.



Elder: "You have a fine life here. Why risk it for nothing?"



Captain: "Not for nothing. For knowing – not for lacking."



Past where cautious sailors turn back, into open water.

Figure 1.1: The decision to leave

The terms of lean and agility are now widely overused. In a previous book I coauthored, great details have been provided to explain the underlying principles of lean and agile software development, how to define a suitable lean development

process, and how to introduce it inside a company.

Still ...

1.1 If we were good enough...

Still, people continue to view agility and lean under a very peculiar perspective. None objects that they are effective, but still most people perceive them as if they are somewhat suboptimal with respect to a more traditional and "scientific" way of managing development processes. The conventional wisdom is that we are lean and agile when/because we are not good enough. For instance, with reference to eXtreme Programming, we are not good enough in planning, so we resort to the planning game. We are not good enough in formalizing requirements, so we adopt the user stories. We are not good enough in documenting, so we try to write the code better so it is self-documentable. But the underlying idea is that if we were good enough, all of this would not happen. If we were good enough, we would write very accurate Gantt charts predicting and describing the overall evolution of development projects, we would write very precise requirements using formal languages, we would document the code effectively and accurately, we would prove the code to the extent possible by the latest advances in formal verification and testing. Moreover, we would communicate effectively with customers, we would secure upfront all the resources needed, we would avoid duplications in the code base, we would write every functionality only once. It is unfortunate, that none has ever been able to do all these beautiful things in the context of a real...

In addition, there is a tendency to apply lean and agility in a very un-lean and un-agile way. Often, people "buy" lean and agile approaches as if they were traditional methods. While it is perfectly self-consistent to adopt a pre-defined traditional methodology as is and to get the organisation or the team to follow religiously its prescriptions, in the sense that such adoption is not contradictory with the notion of the methodology, thinking to just "buy in" a lean or agile method contrast the notion itself of lean and agility, since one of their key premises is to adapt the process to the contest, as it is said in the Agile Manifesto [5] "Responding to change over following a plan."

1.2 Reasons of a controversy

A key reason for controversy is that many women, men, even if philosophers and scientists have been doped by the very alluring ideas of "giving order" to the human life, of organizing or systematising all human activities according to founding principles that are almost mechanical. Works like the "Myth of the Cave" discussed in

the Republic of Plato [34], the "Monadology" of Leibniz [22], or the "Phenomenology of Spirit" of Hegel [15] give a clear evidence of such phenomenon. Moreover, the self-defined *scientific* management coming from Taylor and its application by Ford have further re-empasized such idea. These works give the impression that the any organization or group of people will be saved by the perfect, almost mathematical structure of the activities, by an amazing division of labour where everyone has a task and accomplishes it in details to the uttermost level of perfection, with all tasks together contributing the overall goal, like the different parts of a swiss watch. It is as if these such ordering will give immortality to the human being.



Figure 1.2: Prisoners chained before the shadows on the cave wall, while one climbs toward the light outside.

Books related to the "Project Management Body of Knowledge", like [1], or some prescriptions on how to apply the Capability Maturity Model [16] or the Personal Software Process in software development [18] may have institutionalized this ideology, inadvertently and without any specific intention of the authors in this direction, moving it to classrooms and training generations of students of it as it it were a founding axiom of any organization.

We know how failing such beliefs are. Remaining at the level of philosophers, Nietzsche makes a nice discussion of how societies have managed their organization through well ordered or chaotic structures in his seminal work on the birth of the tragedy [27].

In the previous work, we discussed what happened to Quintus Fabius Maximus: despite winning repeated battles, the Senate dismissed him because his style looked not honourable, using modern words, not enough scientifically grounded, not good enough, and only the disastrous failures of who replaced him forced the Senate to hire him back [19].

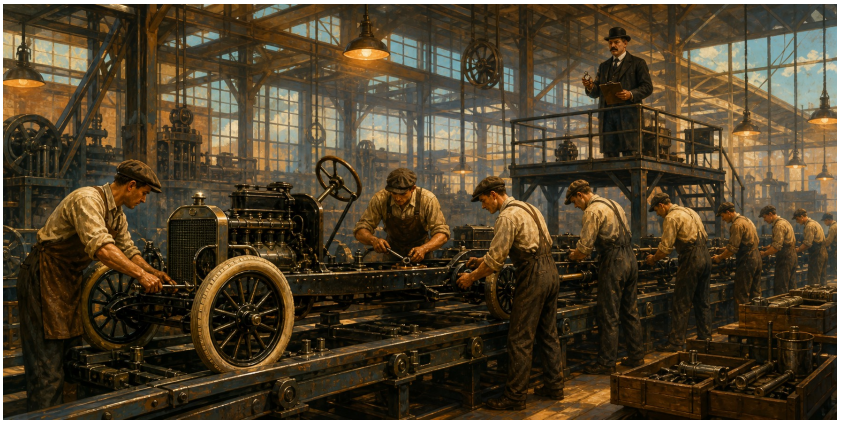


Figure 1.3: Workers perform repetitive tasks on a moving assembly line while an overseer times them with a stopwatch.



Figure 1.4: Apollo, serene and ordered, contrasted with Dionysus, ecstatic and wild, in an ancient amphitheater.



Figure 1.5: Fabius Maximus before the Roman Senate, his victorious army visible behind him, met with suspicion rather than honour.

1.3 Aim of this book

The aim of this book is to go at the root of the principles of lean and agility, to make clear that it is not true that lean and agility are needed because we are not enough good. On the contrary, they come from our comprehensive and careful understanding of how we, as human being think, work, (co)operate, and network. Furthermore, in this book I want to evidence that this understanding is not new; rather it has been a part of the human culture from the very beginning of the civilisation, even though it has been sometimes obfuscated and overshadowed by ideologies of different kinds that claimed to be more scientific, more honorable, better structured, stronger organized simply because they either requiring people to do activities that were sometimes esoteric, required effort or sacrifices, or because they used some sort of mathematical formals or some theoretical framework who gave the impression of a solid formulation, while they were simply superficial and wrong.

Therefore, I review myths, poetry, beliefs, narrations, etc. that are part of our culture. Being Italian, most of the references relate to the Western and Middle Eastern cultures and religions, but this is not exclusive, since a similar wisdom could be found, for instance, in Chinese Taoism, in Indian Vedic and Buddhist thought, or in the oral traditions of Africa and the Americas. This comprehensive perspective should make the reader comfortable of the validity of these principles and on one side will help understanding them deeper and on the other will make the

reader comfortable that using them s/he will not be flagged as not good enough.

Still, I could be objected that the opinion I express are mine and not of the original authors that I cite; to this objection I would reply to a quote I found at a philosophical debate and attributed by Carlo Natali to Umberto Eco in a debate on the modernity of Aristoteles: «a classic is always able to provide new answers»[26].

1.4 (Some of) the founding principles

Now everyone is wondering what these founding principles are. Indeed, a clear, comprehensive, and exclusive definition of them would be contradictory with the tenets of this work. Rather than that, therefore, I list here below some of them in the current

- there are limits of what is knowledgeable and predictable, and not all what is knowledgeable and predictable can be expressed in a rational format; moreover, appreciating such limits does not mean to give up planning but helps planning better and more effectively;
- institutionalizing a behaviour is promoted by first sharing values and principles, and then proposing changes of practices; sudden changes of practices not anchored on values can be promoted by "heroic" actions or witchcraft, but are not going to last long; on the contrary, changes founded on values may trigger permanent changes;
- the manager and the leader in a knowledge based organization is not an absolute dictator, but is someone able to understand the flow of ideas of her/his people and direct them harmoniously to the accomplishment of the goals of the organization; this does not mean that her/his value is lower, actually it ended up being higher because it the result of an act of belief in shared values
- software development is an act of creation, which cannot be systematized even if it can be *somehow* managed and improved
- semeiotic: as old as Hippocrates, this is the discipline of reading the signs a system or a team actually offers – a recurring bug, a friction between two colleagues, a smell in the code – as symptoms of some deeper condition still to be diagnosed, rather than treating them as the disease itself, or worse, explaining them away through an unquestionable doctrine that has stopped being surprised by its own evidence.

1.5 A caveat or two

To avoid any possible misunderstanding, let me make clear again, once, and for all, that this work does not advocate randomness in software development (and knowledge management) organizations, quite like lean and agile processes do not advocate ad-hoc software development processses. On the contrary, it advocates a deeper, more involved and more difficult structures of organizations that requires a more thorough understanding of the human nature, quite like again lean and agile processes require a deeper commitments from managers and developers.

In the book several excerpts from well known works will be presented. To this point it is important to remember that they serve the purpose of evidencing some concept presented in this book, not necessarily the original purpose of the original author. Sometimes, their content is also intentionally stretched to make clear some points considered relevant for this book.

Moreover, to gather a better understanding of them, some of their content is presented both in an English translation and in their original version; nowadays many people speak multiple languages, moreover, there is wide availability of automatic translators – therefore interested readers could try to gather a better understanding of them by scanning through their original version. The provided English translation is the one cited in the reference, to which, though, especially in the case of Dante and of Plato, the author has made adaptations in the hope to make the text more adherent to the original meaning for readers of today.

2
Knowing and Planning

In which the captain proves himself an honest liar.

(Captain Lucian)



Before the tale begins, the captain makes one thing clear.



Rival storytellers: "Every word of it happened!" / "I swear it on my own eyes!"



Captain: κἂν ἓν γὰρ δὴ τοῦτο ἀληθεύσω λέγων ὅτι ψεύδομαι. — "At least I should tell you in full truth, that I am lying." [37, 38]



He sails on — the one honest liar in a sea of dishonest truth-tellers.

Figure 2.1: The honest liar

Science is all about knowledge. Actually, all life is about knowledge – it was in the past and it is today, in the so-called knowledge society. Knowledge has been our tool to conquer the world and to try to make it a better place to live. What is a better device therefore for a scientist or a consultant than to assert knowledge? Very few scientists have been hired for asserting ignorance. Perhaps, to the reader has happened to go to job interviews and to be asked for questions for which there is not a reasonable answer; still, also in these case, it is better to find a kind of an answer, because the interviewer would be disappointed by an assertion of ignorance, assertion that would be very scientific, by the way.

The presumption of knowledge and, even worse, to be able to predict, has affected badly software engineering. Still it is hard for people to accept as a strong and reliable advice to accept the lack of knowledge and act accordingly. Understanding the lack of knowledge is not per se bad, it is an accomplishment, since we can devise action to complement such lack, as we will discuss in section 2.4.

To the knowledge of the writer only Socrates claimed explicitly his ignorance as the cornerstone of his reasoning[1].

2.1 The apology of Socrates

To give a very simplistic view of what happened, the people of Athens were jealous and/or suspicious of Socrates because he was teaching his students to be always a bit skeptical of the principles they were taught and was becoming very popular when he was questioning the existing beliefs. Therefore, he was taken to court with the accusation of corrupting the youth. His defence has become a book by Plato called the "Apology of Socrates" that here we present in the original [33] and in the translated [45] versions.

2.1.1 The words of Socrates

To defend himself from the accusations, Socrates reconstructs why he thinks he was accused and explains how he first disappointed a fellow citizen and then he walked away mumbling.

2.1.2 The unforgivable sin of Socrates

Committing a mistake is and was acceptable for the fellow citizens of Socrates, but admitting ignorance and inability to know and to learn was not. Socrates was sentenced to death for corrupting the youth, and in a sense he was really corrupting

[1] To be precise, this is how Plato perceives and interprets the words of Socrates, since Socrates has never written any work

Original text [33]	**English translation** [45]
πρὸς ἐμαυτὸν δ'οὖν ἀπιὼν ἐλογιζόμην ὅτι τούτου μὲν τοῦἀνθρώπου ἐγὼ σοφώτερός εἰμι: κινδυνεύει μὲν γὰρ ἡμῶν οὐδέτερος οὐδὲν καλὸν κἀγαθὸν εἰδέναι, ἀλλ' οὗτος μὲν οἴεταί τι εἰδέναι οὐκ εἰδώς, ἐγὼ δέ, ὥσπερ οὖν οὐκ οἶδα, οὐδὲ οἴομαι: ἔοικα γοῦν τούτου γε σμικρῷτινι αὐτῷτούτῳσοφώτερος εἶναι, ὅτι ἃμὴοἶδα οὐδὲ οἴομαι εἰδέναι.	For my part, as I went away, I reasoned with regard to myself: «I am wiser than this human being. For probably neither of us knows anything noble and good, but he supposes he knows something when he does not know, while I, just as I do not know, do not even suppose that I do. I am likely to be a little bit wiser than he in this very thing: that whatever I do not know, I do not even suppose I know.»

Quote 2.1: **Apology of Socrates**, 21d

them, because he was breaking down the axioms on which their education was based accepting that there are limits to the human knowledge.

Socrates went even further, deciding to die and not to run away, even if the plan for his escape was ready and safe. The government of Athens would have happily accepted that Socrates would have flown away, since they would have avoided to create a martyr and they would have proven the truth of their accusation.

However, Socrates claimed that he could not go against the law. This is an apparent paradox, on one side he admits the limit of the knowledge and dies because he asserts the importance of knowing that one does not know, and on the other he accept a system based on such flawed knowledge. However, it is not. It is the supreme acknowledgement of the limit of the human knowledge and reasoning. Not only the highest knowledge is to know that we do not know, but, since our knowledge is limited, we cannot do anything but accepting it even when it is absurd and takes us to die.

Therefore, Socrates (or Plato) clearly admit that our knowledge is limited, but also reaffirm that such limitation should not take to shortcuts like superstition. Rather, it should be accepted, still acknowledging that knowledge is the best (and not perfect) way to understand and handle our world.

2.2 Dante

Dante Alighieri (formally Durante di Alighiero degli Alighieri, 1265-1321) is an Italian poet, politician, linguist, and scientist who lived in the Middle Age. There is a common belief that the Middle Age was a period of darkness. This belief was generated at the time of the Renaissance and it became part of the common wisdom, but it is false, especially with reference to the time of Dante [28]. Dante



Figure 2.2: Socrates stands calmly before the Athenian jurors during his trial.

is a notable expression of the rich, diverse, and multicultural Middle Age culture[2].

Dante enjoys studying and discovering new scientific phenomena, exploring the roots of the languages, writing of the love he has toward beautiful women, especially Beatrix.

The teaching of Socrates is fully absorbed by Dante. He has no problem of admitting his ignorance. Still, Dante aspiration of knowledge is sterminate and it is the centrepiece of his biggest and most well known work, the Divine Comedy (or, simply, Comedy, as was called before Boccaccio). In this work, Dante images himself initially lost, perhaps in a middle age crisis, and starts a path to his own improvement. For Dante God is expression of infinite knowledge, a divine knowledge that fully includes the human knowledge. Hence Dante is guided in his itinerary to the full knowledge by who he considered the master of the human, rational knowledge, Vergilius, a pagan poet. Vergilius, and the human knowledge, can take Dante to the full personal purification, and only at that point is not any more enough. Now once this form of knowledge is exhausted Dante reverts to a second form of knowledge, love. That love is a form of knowledge is not an invention of Dante. In the Old Testament making love to someone is often referred to knowing, and not for discretion, but because it is consider the way to get closest to a person. But this is not the just love Dante is referring to. For Dante the human love is an act of aspiration that involve physically and spiritually

[2]A clear example of this society is the habit of some of the women of Florence to go topless, habit that is criticised by Dante in the Divine Comedy, Purgatorio, XXIII, 100-102 [2, 23]. In how many cities would be allowed nowadays to women to go topless?

and that elevates the person to a better understanding of the world and of God. Still this is what is humanly understandable. But is not everything. There is part of knowledge that will never be achievable by a person, neither rationally through deductions nor through love.

Therefore, Dante makes very clear everywhere that on one side his knowledge is limited, it is not possible for it to know everything, and on the other side that not all what is knowledgeable is so though logic reasoning. Something is knowledgeable only through love. He writes in the Divine Comedy:

Original text [2]	**English translation** [23]
State contenti, umana gente, al quia;	Mortals, remain contented at the Quia;
ché, se potuto aveste veder tutto,	For if ye had been able to see all,
mestier non era parturir Maria;	No need there were for Mary to give birth;

Quote 2.2: **Purgatorio** III, 37-39



Figure 2.3: Souls rest contentedly on a terrace of Purgatorio, accepting the limits of human knowledge.

So, it is important not only to understand that we do not know as for Socrates, but also what is knowledgeable rationally and not rationally. Moreover, it is interesting to describe what are for Dante the results of going beyond what is humanly knowledgeable. It looks really like the fate of a software project made by expert

software engineers that loose the ability to understand their limits. Ignoring the limitation of knowledge triggers human failures, big failures that jeopardize what is most valuable in life and sometimes the life itself.

2.2.1 Dante and Ulysses

This is Dante's Ulysses, a very different figure from the narrator of Lucian's tale framing these chapters' comics, though both share a fatal thirst to sail past the known world. Altogether, in the same way in which the aspiration to know what is knowledgeable carries a better life, love, and ultimately gets a person close to God, not understanding the limits breaks all of this. Dante describes this really effectively presenting the story of Ulysses after his come back home at the end of the Troyan war.

Original text [2]	**English translation** [23]
Quando	When I
mi dipartì da Circe, che sottrasse	From Circe had departed, who concealed me
me più d'un anno là presso a Gaeta,	More than a year there near unto Gaeta,
prima che sì Enëa la nomasse,	Or ever yet Aenas named it so,
né dolcezza di figlio, né la pieta	Nor tenderness for my son, nor affection
del vecchio padre, né 'l debito amore	For my old father, nor the due love
lo qual dovea Penelopè far lieta,	Which joyous should have made Penelope,
vincer potero dentro a me l'ardore	Could overcome within me the desire
ch'i' ebbi a divenir del mondo esperto	I had to be experienced of the world,
e de li vizi umani e del valore;	And of the vice and virtue of mankind;
ma misi me per l'alto mare aperto	But I put forth on the high open sea
sol con un legno e con quella compagna	With one sole ship, and that small company
picciola da la qual non fui diserto.	By which I never had deserted been.

Quote 2.3: **Inferno**, XXVI, 90-102

After 10 years of the Troyan war Ulysses come back home. The last step was with Circe, who kept him with her with all sort of seductions, which at the end were overcome by Ulysses. But now, after settling and regaining power there is something still burning in his soul. Something deep, his desire to know, to know what is yet to be discovered, to know the irrational.

What is the highest form of human knowledge for Dante, love, cannot manage to keep him away from this endavour. Love was stronger than seduction, the deep

love for the wife was stronger than the erotic love for Circe, but not stronger than this deepest desire.

So, Ulysses starts a new adventure and gather around him his soulmates – also this part is remarkable, still this act of crazy knowledge was not only an individual act but a communitarian act, a shared act.

2.2.2 The fate of Ulysses according to Dante

Original text [2]	**English translation** [23]
L'un lito e l'altro vidi infin la Spagna, fin nel Morrocco, e l'isola d'i Sardi, e l'altre che quel mare intorno bagna.	Both of the shores I saw as far as Spain, Far as Morocco. and the isle of Sardes, And the others which that sea bathes round about.
Io e ' compagni eravam vecchi e tardi quando venimmo a quella foce stretta dov'Ercule segnò li suoi riguardi	I and my company were old and slow When at that narrow passage we arrived Where Hercules his landmarks set as signals,
acciò che l'uom più oltre non si metta; da la man destra mi lasciai Sibilia, da l'altra già m'avea lasciata Setta.	That man no farther onward should adventure. On the right hand behind me left I Seville, And on the other already had left Ceuta.

Quote 2.4: **Inferno**, XXVI, 103-111

Ulysses sails across the Mediterranean sea and reaches the so-called Hercules Columns, the promontory in front of the Strait of Gibraltar, which defines the area of the what can be known by the human being and then starts acting irrationally. He acknowledges that he and his soulmates are old now. Moreover, he is fully aware the common wisdom that going beyond that point is extremely dangerous, since the streams in the Atlantic ocean are almost deadly threatens to the ships of the time. So the desire of knowledge makes him to perform an act against that logic that should govern knowledge.

Still, he decides to proceed. And to proceed not alone, but with his fellow citizens. To ensure their support he gives a very effective speech.

The speech is a beautiful combination of rhetoric and emotions that thrills the souls of the companies, to the point that they get fully energized. Earlier, Ulysses mentioned that the people on the ship were old and slow, now, after the speech



Figure 2.4: Ulysses and his aged crew row past the Pillars of Hercules as a storm gathers ahead.

they "of the oars made wings for our mad flight." He moves the hearts of his mates. He evokes their roots and their ancestral desire to know and to know more.

It is beautiful how Dante describes how a group of people is lured and convinced by an amazingly attractive leader. And if is enchanting the striking contradiction in the words of Ulysses: he tells to his mates that they "were not made to live like unto brutes but for pursuing virtue and knowledge." and in saying so uses what is most in contrast with virtue and knowledge, seduction and misinterpretation of the reality. Virtue would have required them to stay with they loved ones, to care of their parents, children, wives. Knowledge would have required them to understand that the endavour was impossible, as Ulysses himself knew. Still he could not resist the temptation and they could not.

Earlier, as described in the Odyssey, Ulysses wanted to experience the unexperienceable, the singing of the mermaids. But to do so, he asked his people to bind him to a pole so he could not move and neutralised all his mates blocking their ears, so they could not hear the deadly songs. Now, he does not do this any more, on the contrary deliberately decide to proceed. And his destiny is sealed.

Ulysses travels for 5 months. During these days he gathers knowledge of the surrounding environment[3]. There is no mention of any tiredness of the crew, so apparently they were still going at full speed after 5 months. Then, something unexpected happens. They see a mountain and think that finally they have found the land. But something like a terrible whirlwind appears in the air, comes from

[3]For those interested, indeed Dante was quite sure that the earth was a ball and not a dish and that the stars that appeared in the southern hemisphere were not necessarily the same as the one in the northern.

Original text [2]	**English translation** [23]
"O frati," dissi, "che per cento milia	'O brothers, who amid a hundred thousand
perigli siete giunti a l'occidente,	Perils,' I said, 'have come unto the West,
a questa tanto picciola vigilia	To this so inconsiderable vigil
d'i nostri sensi ch'è del rimanente	Which is remaining of your senses still
non vogliate negar l'esperïenza,	Be ye unwilling to deny the knowledge,
di retro al sol, del mondo sanza gente.	Following the sun, of the unpeopled world.
Considerate la vostra semenza:	Consider the seed from which you came from;
fatti non foste a viver come bruti,	You were not made to live like unto brutes,
ma per seguir virtute e canoscenza".	But for pursuing virtue and knowledge.'
Li miei compagni fec'io sì aguti,	So eager did I render my companions,
con questa orazion picciola, al cammino,	With this brief exhortation, for the voyage,
che a pena poscia li avrei ritenuti;	That then I hardly could have held them back.
e volta nostra poppa nel mattino,	And having turned our stern unto the morning,
de' remi facemmo ali al folle volo,	We of the oars made wings for our mad flight,
sempre acquistando dal lato mancino.	Evermore gaining on the larboard side.

Quote 2.5: **Inferno**, XXVI, 112-126

the mountain, and destroys the ship. There is no hope of survival for them.

Only at this point Ulysses acknowledges that there are things outside his control. Describing the whirlwind he says that it destroys the ship "as pleased Another," meaning "as God wanted." At this point there is no more hope, as at the end "the sea above us closed again."

2.2.3 Dante on Ulysses

There is no need to emphasize that Dante really likes the figure of Ulysses. He is attracted by him and sees in him his own personality, traits, and especially his desire to know. Still, it is firm in evidencing his human, very human mistake, his inability to acknowledge the unknowledgeable, his arrogance in thinking to be able to handle all possible situations, simply by properly planning and energizing his crowd.

In particular, this trip of Ulysses has to been seen in perspective of his earlier trip from Troy back to Ithaca. The earlier trip was also incredibly difficult. it was a "wicked project," using the terms of Lean Software Development [19], a situation where there is no analytical solution. But the trip back was guided by a true desire for knowledge and love, a legitimate desire, not meaningless personal ambitions.

Original text [2]	**English translation** [23]
Tutte le stelle già de l'altro polo	Already all the stars of the other pole
vedea la notte, e 'l nostro tanto basso,	The night beheld, and ours so very low
che non surgëa fuor del marin suolo.	It did not rise above the ocean floor.
Cinque volte racceso e tante casso	Five times rekindled and as many quenched
lo lume era di sotto da la luna,	Had been the splendour underneath the moon,
poi che 'ntrati eravam ne l'alto passo,	Since we had entered into the deep pass,
quando n'apparve una montagna, bruna	When there appeared to us a mountain, dim
per la distanza, e parvemi alta tanto	From distance, and it seemed to me so high
quanto veduta non avëea alcuna.	As I had never any one beheld.
Noi ci allegrammo, e tosto tornò in pianto;	Joyful were we, and soon it turned to weeping;
ché de la nova terra un turbo nacque	For out of the new land a whirlwind rose,
e percosse del legno il primo canto.	And smote upon the fore part of the ship.
Tre volte il fé girar con tutte l'acque;	Three times it made her whirl with all the waters,
a la quarta levar la poppa in suso	At the fourth time it made the stern uplift,
e la prora ire in giù, com'altrui piacque,	And the prow downward go, as pleased Another,
infin che 'l mar fu sovra noi richiuso.	Until the sea above us closed again.

Quote 2.6: **Inferno**, XXVI, 127-142

This trip, in converse, is almost doped. We can go back to the earlier words to stress this:

Original text [2]	**English translation** [23]
né dolcezza di figlio, né la pieta	Nor tenderness for my son, nor affection
del vecchio padre, né 'l debito amore	For my old father, nor the due love
lo qual dovea Penelopè far lieta,	Which joyous should have made Penelope,

Quote 2.7: **Inferno**, XXVI, 75-77

Let me emphasize again "Nor tenderness for my son, nor affection for my old father, nor the due love, which joyous should have made Penelope," none of this stopped him. He was blind and drugged by himself and by his narcissism to the point of loosing rationality. And, again, as mentioned above the speech of Ulysses is an example of such doping, so beautiful, but so intrinsically crazily contradictory: planning the unplanned, claiming irrationally to pursue knowledge and science against all rules of rationality. Doping has a short term effect, as we will see again in Chapter 3, people work very hard, become supermen, fly, go against age. But

this cannot succeed. There is a failure at the end. The failure of the man who goes after idols (also this point will be expanded further in Chapter 3).

The reader should be aware, though, that this human failure is not the reason why Ulysses went to hell. This was a human failure, not a sin. The sin of Ulysses is other, even if driven by the same desire. He has devised the Troyan horse, gambling the Troyan people with the belief that the wooden horse was coming from a god. Ulysses has won the war, but dishonestly, setting his own rules of beheviour. Therefore, in Troy he has committed a bad sin, playing with the good faith of people to kill them. In a sense he has felt like being God – quite the same situation as in the case of this trip. So in this Dante wants to emphasize this strict connection, the inability of Ulysses to stay within the rules, the rules of rationality or the rules of God (which for Dante are compatible, even if not always explicable, with rationality), his lack of perception of his limits drive him to the human and spiritual failures.

2.3 Sunter and Berlin: Archilochus and Tolstoy

Clem Sunter gave an inspiring keynote at ICSE 2010 based on the seminal work of Isaiah Berlin titled "The Hedgehog and the Fox: An Essay on Tolstoy's View of History" [6]. He started with the famous verses of a poetry by Archilochus, vastly interpreted throughout the centuries and also the starting point of the work of Berlin [6].

Original text [3] as quoted in [6]	**English translation** in [6]
πόλλ' οἶδ' ἀλώπηξ	The fox knows many things,
ἀλλ' ἐχῖνος ἕν μέγα	but the hedgehog knows one big thing

Quote 2.8: **Archilochus fragment 201**

Sunter (and Berlin) use these verses as a way to divide the human thinkers and historians. The fox is the metaphor of people who accept the impossibility of defining a single uniform rule of behaviour and view the world and its phenomena as a set of interconnected entities, mutually interacting, often in a random and completely unstructured way. The fox thus behaves trying to sense the environment, and to experiment possible actions until it finds the way to achieve its goal.

The hedgehog, on converse, is the metaphor of those people who believe that there is such unifying rule and try to explain all the phenomena accordingly, moving strictly according to the plan, in a very protected and cautious way.



Figure 2.5: A fox and a hedgehog face each other on a quiet path at dusk, after Archilochus's distinction as recovered by Berlin [6].

Aristotle, Ariosto, Kant, and Bergson belong by all means to the foxes while Plato, Petrarca, Hegel, and Nietzsche to the second. The reader may identify people who to varies degrees are hedgehog or foxes in their network, among politicians, and historical figures.

What is really peculiar of Tolstoy, according to Sunter and Berlin is that he claims to hedgehog but in reality he is a fox. In all his work, and especially in the second part of his life, Tolstoy tries to find an order, a unifying rules among the things that happen and the activities that we do, but still he admit the impossibility of finding such overarching rule. Most situations occur just by chance or according to unexplainable dynamics, as Berlin argues in the following excerpt.

English text [6]

With great force he argues that only those orders or decisions issued by the commanders now seem particularly crucial (and are concentrated upon by historians) which happened to coincide with what later actually occurred; whereas a great many other exactly similar, perfectly good orders and decisions, which seemed no less crucial and vital to those who were issuing them at the time, are forgotten because, having been foiled by unfavourable turns of events, they were not, because they could not be, carried out, and for this reason now seem historically unimportant.

Quote 2.9: **The Hedgehog and the Fox**, II

Berlin looks like some empirical software engineer discussing the difficulty of

generalizing rules from specific events. He is also so similar to us who try to put order on the things we do, but acknowledge the impossibility of doing it in a complete and systematic way, and we accept that success may just come because of chance, good luck, or some other reason, indeed existing but not (perhaps yet) understandable.

2.3.1 On the possibility to explain the sequences of events

The discussion becomes even more interesting analysing the perception of the uncertainty of the evolution of the humankind and of the sequences of events present in Tolstoy, an uncertainty that surrounds any human endavour, and, among the other (we claim) software development. Berlin writes as follows.

English text [6]

The new history is like a deaf man replying to questions which nobody puts to him . . . the primary question . . . is, what power is it that moves the destinies of peoples? . . . History seems to presuppose that this power can be taken for granted, and is familiar to everyone, but, in spite of every wish to admit that this power is familiar to us, anyone who has read a great many historical works cannot help doubting whether this power, which different historians understand in different ways, is in fact so completely familiar to everyone.

Quote 2.10: **The Hedgehog and the Fox**, III

Tolstoy writes on historical facts and analyses how success was achieved and tries to determine the key factors that lead to a situations, like victory. He wants to understand "what power is it that moves the destinies of peoples," what is the underlying logic that can explain the life of the humankind, and he comments as follows.

Here, the language is a bit hermetic. Tolstoy calls the reality "modern history," since this is the universe of his analysis, and the sequences of events "questions" and "answers," as if there is a causality, like the one occurring between a question and an answer to such question. He does not deny the existence of a logic underlying the reality. However, this logic produces sequences of events, "questions" and "answers," whose connections are not understandable, "modern history, like a deaf man, answers questions no one has asked."

Altogether, Tolstoy is not calling for a mystic interpretation of a reality; rather, he is acknowledging the impossibility to understand, still within the deep desire of understanding. This is even more clear in the next excerpt.

Altogether, Tolstoy is not calling for a mystic interpretation of a reality; rather, he acknowledges:

Original text [41]	**English translation** [42]
Странность и комизм этих ответов вытекают из того, что новая история подобна глухому человеку, отвечающему на вопросы, которых никто ему не делает. Если цель истории есть описание движения человечества и народов, то первый вопрос, без ответа на который все остальное непонятно, – следующий: какая сила движет народами?	The strangeness and absurdity of these replies arise from the fact that modern history, like a deaf man, answers questions no one has asked. If the purpose of history be to give a description of the movement of humanity and of the peoples, the first question – in the absence of a reply to which all the rest will be incomprehensible – is: what is the power that moves peoples?

Quote 2.11: **Second Epilogue**, I

Original text [41]	**English translation** [42]
История как будто предполагает, что сила эта сама собой разумеется и всем известна. Но, несмотря на все желание признать эту новую силу известною, тот, кто прочтет очень много исторических сочинений, невольно усомнится в том, чтобы новая сила эта, различно понимаемая самими историками, была всем совершенно известна.	History seems to assume that this force is self-evident and known to everyone. But in spite of every desire to regard it as known, anyone reading many historical works cannot help doubting whether this new force, so variously understood by the historians themselves, is really quite well known to everybody.

Quote 2.12: **Second Epilogue**, I

- the existence of a logic in the history and in the sequence of events, logic that appears simple to understand;
- the desire of the people to understand such logic and then to use such understanding to model their reality and to plan their future actions to accomplish their goal;
- the practical impossibility of understanding it.

This is why at the end Tolstoy and Berlin advocate the behaviours of foxes, rather than of hedgehogs.

How much software engineering is in these statements and excerpts! How many time we have desired and thought to understand our ongoing projects, that appeared so easy to understand. How many time we have modelled them, for instance using Gantt, Pert, and so on. And how many time we have then miserably failed because the flow of event twisted in a totally inexplicable way.

2.3.2 The traditional software engineer

In the context of such uncertain situation, a typical situation is well described by Berlin in the following comment on War and Peace.

English text [6]

In a famous passage dealing with the state of Moscow in 1812 Tolstoy observes that from the heroic achievements of Russia after the burning of Moscow one might infer that its inhabitants were absorbed entirely in acts of self-sacrifice – in saving their country or in lamenting its destruction, in heroism, martyrdom, despair – but that in fact this was not so. People were preoccupied by personal interests. Those who went about their ordinary business without feeling heroic emotions or thinking that they were actors upon the well-lighted stage of history were the most useful to their country and community, while those who tried to grasp the general course of events and wanted to take part in history, those who performed acts of incredible self-sacrifice or heroism, and participated in great events, were the most useless.[4] Worst of all, in Tolstoy's eyes, were those unceasing talkers who accused one another of the kind of thing «for which no one could in fact have been responsible»; and this because «nowhere is the commandment not to taste of the fruit of the tree of knowledge so clearly written as in the course of history. Only unconscious activity bears fruit, and the individual who plays a part in historical events never understands their significance. If he attempts to understand them, he is struck with sterility.»[5] To try to 'understand' anything by rational means is to make sure of failure.

Quote 2.13: **The Hedgehog and the Fox**, III



Figure 2.6: Moscow burning in 1812: an ordinary woman and cart-driver go about their business unnoticed, while self-styled heroes strike dramatic poses and accusers point fingers at one another, after Tolstoy as read by Berlin [6].

The situation is dramatic, the standard rules of behaviours are gone. In this context there are people who try to survive, the foxes: they focus on the short term goal, acknowledging that any plan for a long term goal is unfeasible – they are the most useful. There are then the people who try to adhere to predefined principles, who try to apply abstract or hypothetic rules, performing act of heroism – they are damaging because there is no more rationality in their actions, just an attempt to anchor there real instability to roles, who, being grounded on pure empirical evidence, do not carry any abstract or generalizable value. Lastly, there are the worst people. The one who want to apply to everyone the wrong assumption on the world and who blame others accusing them to be the reasons of the problem for not having followed strictly the rule.

This is a reflection of our world. A typical behaviour of the "traditional software engineer," the one of Figure 5.6 of [19], is to try to become a zealot, to follow strictly the rules. I think that many time we have experienced this situations – colleagues or managers that if there is a problem or a crises try to find solutions in procedures and protocols. Tolstoy comments that these approaches do not work.

Then, for Tolstoy, there is the worst category of people, the defenders of the faith – the one who start blaming the others for not being precise enough, for not having followed the procedures, not having adhered to the rules, for not having planned with enough precision. They are the most damaging; they appear the ones

who killed Socrates, the ones who appear to think honestly (not to please their pharaohs or bosses) that the rules are their sacred means to achieve immortality, and that breaking such rules destroys forever the possibility of achieving such immortality. Also in software engineering we experiment this worst category of people: we have colleagues who try to indoctrinate us religiously on some more or less minimal aspect of development (like how to indent the code, how to name variables, how to organize files, ...) as if they are explaining some breakthrough discovery of science, and who cannot accept any other position but theirs. Some of the ideas that they advocate might be even generally good, however, they turn such ideas upside down, transforming them from useful practices to overarching principles.

Now the key question, still with reference to Figure 5.6 of [19] is whether Tolstoy is endorsing not only agile software engineers but also cowboy coders – after all who are those "who went about their ordinary business?" Well, cowboy coders can be as dogmatic as traditional software engineers in their belief. Cowboy coding does not simply mean a chaotic approach to software production, but also means the inability to enact sound mechanisms to focus on value – cowboy coders do not go "about their ordinary business," rather they focus in the personal fulfilment regardless of the business they are in.

2.4 What do we do once we assert ignorance or inability to predict

Altogether, throughout this chapter we have reemphasized that:

- there is an intrinsic limitation to the human ability to know and to predict facts and events;
- not all the knowledge can be expressed in rational form; indeed, the one that can is an even smaller subset; in particular, it is Beatrix that takes Dante to the discovery of God, so love for a woman can make Dante understand what he did not understand rationally, but still not everything, meaning that not always knowledge is or can be expressed in formal rational forms, something very common in software and knowledge based projects;
- people throughout the history have presented and discussed such limitation; acting as if it did not exist does not demonstrate smartness, in the best case evidences an uncontrollable and unstoppable desire to go beyond the well established rules (as in the case of Dante/Ulysses), usually it is a symptom of superficiality and ignorance (as in the case of Plato/Socrates);

- moreover, acting as if it did not exist, typically result in personal, human, and business failures.

Ok, so what? someone could ask me.

First of all, we have already made a big point. Receiving a "I do not know!" or a "I cannot predict!" as an answer in any project and especially in a software engineering project is not necessarily a sign of laziness or inability to manage, sometimes it is exactly the opposite.

The second point is that indeed we can do something. Again we resort to the words of Dante.

Original text [2]	**English translation** [23]
Quel sol che pria d'amor mi scaldò 'l petto,	That sun, which erst with love my heart warmed,
di bella verità m'aveva scoverto,	Of the beautiful truth had unto me discovered,
provando e riprovando, il dolce aspetto;	By proving and reproving, the sweet aspect.

Quote 2.14: **Paradiso**, III, 1-3

Dante is deeply in love for Beatrice and this love takes him to the highest possible level of perfection that a human mind can reach. It is a love that burns in his heart. Still, the whole truth is achieved via "proving and reproving," not yet full scientific method of Galileo [13], because it was mostly referred to philosophical and theological debates, but still something concrete and empirically verifiable. So the true love, the true emotion, the true desire to understand not only the shallow appearance, but the deep meaning of the facts, is the prerequisite for a deeper understanding of the events that still need a followup argumentation through a dialogic and possibly empirical interaction. This is itself a cornerstone in managing uncertainty, being able to relate suppositions to concrete fact or move away from it.

There is an interesting anecdote about this. Many software engineers might have read and even brainwashed by the famous citation of Tom De Marco "You can't control what you can't measure" [24]. This sentence makes apparently perfect sense in an abstract world. How can I control any process if I cannot at least measure it, see how it grows, how big becomes etc. However, in the reality such abstract statement provides very little practical value, to the point the same Tom De Marco has later completely recanted it [9], stating "So, how do you manage a project without controlling it? Well, you manage the people and control the time and money. You say to your team leads, for example, 'I have a finish date in mind, and I'm not even going to share it with you. When I come in one day and

tell you the project will end in one week, you have to be ready to package up and deliver what you've got as the final product. Your job is to go about the project incrementally, adding pieces to the whole in the order of their relative value, and doing integration and documentation and acceptance testing incrementally as you go."'

Altogether, even Tom De Marco advocates an incremental management, based on empirical analysis of the reality centered on systematic and constant acquisition of knowledge of the surrounding environment, "proving and reproving" every statement we acquire about the process and its evolution.

The third point is more philosophical. It appears that being engaged in an experimental and emerging discipline, like software engineering, has an implication on how should be the overall vision of the world and of the organization of endavours, projects, and activities – at least, those endavours, projects, and activities that are work related... Big visions are indeed important and may help to set the world, but movements as foxes appear to be most effective in coping with an ever-changing and dynamic world, where discoveries sometimes lead to new solutions well beyond their initial intentions (think at the Internet, Unix/Linux, SQL, Facebook itself, ...) and long planned projects fail catastrophically (think at Multics, Apple Newton, Windows ME, IBM PCjr, Oracle CPG , ...).

2.5 So ... what?

Imagine you had to deal with a colleague claiming that the only reason why we cannot find an analytic solution to $(\int \mathrm{e}^{-x^2}\,\mathrm{d}x)$ is our lack of effort in solving it ... what would you say? This is not so difference to what happens to us in many situation of our software engineering life, and the authors hope that this chapter has given you an answer to these situations. Altogether, at the end of this chapter, we list a few take away for a software engineering reader:

- **claim ignorance any time needed**: learning from Socrates, there is no value, no science and no engineering in claiming that we know what we do not know; when asked to make a plan, if a task is unfeasible or it is simply unpredictable, it is much more scientific to tell it very clear rather than trying to formalize it into a nice structure, like a Gantt chart, without any real content; admitting ignorance can be confused for laziness or lack of knowledge, but this is not the case, and, anyway, it is better to be considered lazy or unknowledgeable rather than to promise what is unfeasible;

- **our narcissistic desire for heroism in a project can be a major source of troubles**: in a software project we do not want to replicate the experience of the Ulysses of Dante, we do not want to try the unreasonable

because we are doped by our narcissistic desire of self-pleasing; rather we want to deal with committed people, who put all themselves toward the goal with a clean, transparent, self-aware desire to success; we ought to remember the comment of Tolstoy that sometimes the heroic approach is damaging;

- **in case of doubt, experiment with no prejudices**, as Dante suggested even dealing with the love for Beatrice;
- **beware of the defender of the faith**: they are just the worst people to handle, they do not see the big pictures and are even able to twist your words extracting from some useful advice that you are giving their generalization, and attribuing it to you;
- in general, **be a fox**! Software development is full of uncertainty, the underlying theory is still mostly unstable, the target platforms for the systems being developed are typically work in progress; all of this triggers the impossibility to move like a hedgehog; we need to be able to sense the reality, to perceive the grey areas of lack of knowledge, and to move incrementally in a complex world.

3
Enacting processes

In which they land on an island where comfort takes root – literally, for those who stay too long.

(Captain Lucian)



Landfall on an island where nothing is asked of anyone.



Vine-woman: "Drink. Stay. You need nothing more than this."



Sailor, already rooting: "Just a little longer – this is easier."



The captain pulls the rest back in time — but one sailor stays, rooted, and the ship sails on without him.

Figure 3.1: They ended up on an island where comfort takes root — literally, for those who stay too long.

3.1 Humans and processes

People perform actions to achieve goals. They learn from their early childhood the key survival actions by playing with their parents and with other children. In this humans are very similar to other animals. It is interesting to consider for example cats. Cats need to learn to move fast and to hunt. So the mother cat teaches this behaviour to their children: she forces them to move, she let them play one another, she moves forcing them to follow her, and then she even encourages them to follow little targets as if they were little prays, for instance moving her tail and letting the kitten to chase it.

Likewise, humans learn from experience the actions to undertake and pass them from generation to generation, even if such actions are not scientifically proven. An illustration of this is given by the following letter of Cicero to his servant and friend Tiro. Tiro is sick and away and Cicero gives him suggestions on how to treat his gastritis.

Original text [7]	**English translation**[1]
de medico et tu bene estimari scribis et ego sic audio, sed plane curationes eius non probo. ius enim dandum tibi non fuit cum κακοστόμαχος esses. sed tamen et ad illum scripsi accurate et ad Lysonem.	About the doctor even though you write that you have a good opinion of him and the same good opinion I have heard from other sources, I do not agree with his therapy. Since you suffer from gastritis, you should not eat soups. Anway, I wrote this to him in details and also to Lyso.

Quote 3.1: **Letters to Friends**, XVI, 4

But ... wait a second, Cicero is not a doctor – how can he give advices about health to Tiro, to the point of claiming that a famous and respectable doctor is wrong? But ... wait another second ... at the time of Cicero there were not studies of the digestive system, and the common medicine in use, the one grounded on the theories of Hippocrates, was a mix between very simple empirical findings and some arcane theories of the structure of the human body – for instance, it was believed that the arteries carry air (hence their name).

Still, Cicero, a very rational eclectic, almost agnostic, still grounds his suggestions to a person for which he deeply care on uses based on almost nothing. This is to say that the processes we learn become an integral part of our life and we often believe in them as if they come from God.

One can read this form an anthropological perspective. The human being was created to live in the wilderness, not in cities. The ability to enact processes without too much thinking was an essential survival skill. If you meet a tiger in the middle of the savannah, you do not start thinking at what to do, you simply



Figure 3.2: Cicero writing a letter of advice to his sick friend and secretary Tiro.

enact your ancestral process – you run away. This reaction to fear is reflected also, for instance, in anxious situations: cold sweats and increased heart rate could be interpreted as typical phenomena arising when a (primitive) human needed to escape from a dangerous situation, requiring a higher amount of oxygen in the muscles and the ability to cool down its muscular system to run faster.

Altogether, we, humans, learn processes to react to situations since our early childhood explicitly or also implicitly, and these processes are very difficult to change. Inevitable, these processes affect how we do our work, including software development. It becomes therefore essential that we learn how to define processes and then how to implement them changing how people used to do things.

Under this perspective, it is interesting to consider the XP approach. In XP, the definition of a methodology is based first in outlining a set of common values, then on deriving from values fundamental directions, and only eventually to go from fundamental directions to practices. Initially, one could question the need of such triple partition. After all, developers focus on practices. We will see that actually such approach is very effective way to go.

First, we turn our attention to how the humankind has referred to values, fundamental directions, practices, and their mutual relationships historically and how has reasoned on persuading people of adhering to them, enduring permanent changes of behaviours.

To this end we start referring to a story coming from the Old Testament of the Bible. This story is reported almost in the same way in the Qur'an. Then there will be the reference of the beginning of the Gospel, which is indeed mostly

relevant to the Christianity and partly to Islam. The goal is not to proselitize or to assert the superiority of this or that religion. Rather the purpose is to analyse the essential anthropological meaning of it, quite in the same way done by famous agnostic or atheistic thinkers like Benedetto Croce [8] or Eric Fromm [11]. The readers is invited to reflect from a pure psychological and sociological perspective the reason why the Jewish society, and then also the Christianity and the Islam pose so much importance to this story and the Christianity and, partly, the Islam to its continuation. We want to discuss its underlying dynamics and to evidence how such dynamics affect also the lean and the agile approaches to knowledge (and software) production. To avoid any possible bias we have adopted the original "Vulgata" of Saint Jerome [20] and so-called "Current English Version" of the Bible as is reported on the website http://www.biblija.net as visited on the 8th April 2016 [39].

3.2 The tables of the law – an anthropological view

We refer first to when Moses is given the Table of the Law by God. This is an essential part of the history of Judaism and we can consider that embodies the Jewish wisdom on how to create an organized community of people working toward a goal.

3.2.1 God gives the laws to Moses

The description starts simply with a clear statement that God talks to Moses and gives him the tables with his law. The language is simple and essential.

Original text [20]	**English translation [39]**
Deditque Dominus Moysi, completis huiuscemodi sermonibus in monte Sinai, duas tabulas testimonii lapideas scriptas digito Dei.	When God had finished speaking to Moses on Mount Sinai, he gave him the two flat stones on which he had written all his laws with his own hand.

Quote 3.2: **Exodus** 31, 18

In practice, God defines an agreement with his people setting up a written law. Let's contextualize! Some of you might have seen the movie "The Ten Commandments," featuring Charlton Heston and Yul Brynner. Hollywood might give a quite distorted representation of history, but still the way God or the gods were

pictured to relate to the humans was quite like in the movies. Human sacrifices were common, there were people trying to understand the desired of God(s) and to predict the future looking at how the birds were flying, reading the stars, and in other similar ways. Above all, the divinity was untouchable and unapproachable. The society was an absolute monarchy, where the head had complete and unregulated power of life and death on all his or her people. The Jewish society was a remarkable exception; it was closer to a democracy, using today criteria. The wisdom of the Jew understood that to organize for a long period of time such free society, centered on the individual decision to leave Egypt, written laws were required: neither an adorable or fearful God, nor an inspiring leader were enough. Even more, God gives up some of his secrecy to write himself the laws. This is revolutionary for that time, and this is the root of a civilization that has been lasting thousands of years across the most terrible events that the humankind may remember.

Altogether, in the Jewish approach a change to a new behaviour or process is achieved by having people discussing all at the same level and agreeing very clearly, possibly in writing, on their general values, fundamental directions, and practices. We are assisting at a process where the emotional aspect of the belief is joint to the rational process of coding the belief in written format.

Then, if the act of writing the laws is important, it is almost more interesting to investigate what such laws are and how they were able to reorient, reorganize, and stabilize a society lasting thousands of years.

Initially, the laws assert their value, using as criteria the experience that people had.

Original text [20]	**English translation [39]**
locutus quoque est Dominus cunctos sermones hos. «Ego sum Dominus Deus tuus qui eduxi te de terra Aegypti de domo servitutis.»	And the Lord spoke all these words: «I am the Lord your God, who led you away from the land of Egypt, out of the house of servitude.»

Quote 3.3: **Exodus** 20, 1-2

God claims to be someone who has a proven value, is who has lead the people away from the slavery of Egypt. God does not define him with other, more abstract attributes, but it connects itself with the life of the people.

Now, from the wisdom of the Jewish society, we learn that permanent changes have to be rooted on values, especially on values that are tangible and that people

have experienced, not on abstract and intangible principles.

Then, going ahead in the analysis, God continues by setting a set of paramount fundamental directions: these are the enablers to achieve the key value, the inner approach that one has to have to move in the direction proposed by God. First and foremost, the unicity of him and of his law and the impossibility of having alterations of him, worshipping other gods, or even mentioning them or using the name of God without a good purpose. We can refer these fundamental directions to the desire of clearing the ground from any possible misinterpretation.

Original text [20]	**English translation [39]**
«Non habebis deos alienos coram me. Non facies tibi sculptile neque omnem similitudinem quae est in caelo desuper et quae in terra deorsum nec eorum quae sunt in aquis sub terra. Non adorabis ea neque coles.»	«You shall not have strange gods before me. You shall not make for yourself a graven image, nor a likeness of anything that is in heaven above or on earth below, nor of those things which are in the waters under the earth. You shall not adore them, nor shall you worship them. »
«Ego sum Dominus Deus tuus fortis zelotes visitans iniquitatem patrum in filiis in tertiam et quartam generationem eorum qui oderunt me et faciens misericordiam in milia his qui diligunt me et custodiunt praecepta mea.»	«I am the Lord your God: strong, zealous, visiting the iniquity of the fathers on the sons to the third and fourth generation of those who hate me, and showing mercy to thousands of those who love me and keep my precepts.»
«Non adsumes nomen Domini Dei tui in vanum nec enim habebit insontem Dominus eum qui adsumpserit nomen Domini Dei sui frustra.»	«You shall not take the name of the Lord your God in vain. For the Lord will not hold harmless one who takes the name of the Lord his God falsely.»

Quote 3.4: **Exodus** 20, 3-7

God needs people who understand the key value he is proposing and who share his fundamental directions. Asking not to take the name of the Lord in vain means not to use it for the own purposes, not to bend it to the own values and fundamental directions (not to mention, practices). Here God tells all and everything he intends to say. People should not worry or second guess something else. We have already talked in the previous chapters of the Zealots and of the defenders of the faith. They are not needed in this context.

We are now learning that lasting changes of behaviours are implemented setting after values overarching fundamental directions, which help people focusing on values and remove all possible reinterpretations or second-guessing that people

may have.

Lastly, God specifies the practices.

Original text [20]	**English translation [39]**
«Memento ut diem sabbati sanctifices, sex diebus operaberis et facies omnia opera tua; septimo autem die sabbati Domini Dei tui non facies omne opus tu et filius tuus et filia tua servus tuus et ancilla tua iumentum tuum et advena qui est intra portas tuas. Sex enim diebus fecit Dominus caelum et terram et mare et omnia quae in eis sunt et requievit in die septimo idcirco benedixit Dominus diei sabbati et sanctificavit eum. »	«Remember that you are to sanctify the day of the Sabbath. For six days, you will work and accomplish all your tasks. But the seventh day is the Sabbath of the Lord your God. You shall not do any work in it: you and your son and your daughter, your male servant and your female servant, your beast and the newcomer who is within your gates. For in six days the Lord made heaven and earth, and the sea, and all the things that are in them, and so he rested on the seventh day. For this reason, the Lord has blessed the day of the Sabbath and sanctified it.»
«Honora patrem tuum et matrem tuam ut sis longevus super terram quam Dominus Deus tuus dabit tibi. Non occides. Non moechaberis. Non furtum facies. Non loqueris contra proximum tuum falsum testimonium. Non concupisces domum proximi tui, nec desiderabis uxorem eius, non servum, non ancillam, non bovem, non asinum, nec omnia quae illius sunt.»	«Honor your father and your mother, so that you may have a long life upon the land, which the Lord your God will give to you. You shall not murder. You shall not commit adultery. You shall not steal. You shall not speak false testimony against your neighbor. You shall not covet the house of your neighbor; neither shall you desire his wife, nor male servant, nor female servant, nor ox, nor donkey, nor anything that is his.»
Cunctus autem populus videbat voces et lampadas et sonitum bucinae montemque fumantem et perterriti ac pavore concussi steterunt procul.	Then all the people considered the voices, and the lights, and the sound of the trumpet, and the smoking mountain.

Quote 3.5: **Exodus** 20, 8-18

We would like to remark three aspects. The first practice refers to the dignity of people – God compares them to him, and in the same way he rested after 6 days of work, the same should the people do; to this respect, Eric Fromm makes an extensive discussion in his book *To Have or to Be* [10], excerpted below.

The second point to remark is that all the other practices are very clear to follow and not ambiguous, not leaving any part to interpretation.

English text [10]

The Shabbat is the most important of the biblical concepts, and later of Judaism. It is the only strictly religious command in the Ten Commandments: its fulfilment is insisted upon by the otherwise antiritualistic prophets; it was a most strictly observed commandment throughout 2000 years of Diaspora life, wherein its observation often was hard and difficult. ... In order to understand this role we must penetrate to the core of the Shabbat institution. It is not rest per se, in the sense of not making an effort, physically or mentally. It is rest in the sense of the re-establishment of complete harmony between human beings and between them and nature. Nothing must be destroyed and nothing be built: the Shabbat is a day of truce in the human battle with the world. ... On the Shabbat one lives as if one has nothing, pursuing no aim except being, that is, expressing one's essential powers: praying, studying, eating, drinking, singing, making love.

Quote 3.6: **To Have or to Be**, III

The third point is that everything terminates again with an act of rational knowledge acquisition: "populus videbat voces," in English "people considered the voices," where the verb "considered" is a somehow weak translation of the latin term "videbat", which could also be translated as seeing and as perceiving. So the people rationally acquired the law that God gave them in written format.

3.2.2 On the workflow followed by God in structuring the Ten Commandments

A brief summary of the overall workflow can be interesting to understand the structure of this fundamental step in the history of the Jewish civilization.

- God gives the laws in written format, committing himself to a rational definition of his desire in a stable format, understandable by human.
- Initially, the laws state the overarching values of themselves, rooting their value on the personal experience of people receiving them.
- Then, the laws handles the fundamental directions underneath them, the first of it being, their unicity and the ban for any possible misinterpretation.
- Finally, the laws define a set of practices, the first being a tribute to the dignity of the human beings that are following such laws.
- Everything terminates with an explicit act of rational knowledge.

This is how the people of Israel managed to reorient their activities, perform a change in their practices, and, overall, enact a new process.



Figure 3.3: Moses descending Mount Sinai with the tablets of the law, carrying order down from the mountain to the camp below.

3.2.3 On the workflow followed by Kent Beck in structuring XP (first way)

Indeed, this initial narration gives us a lot to think about. What is immediately striking is the analogy with how Kent Beck presents XP, especially in his early work [4], where he describes his methodology with three steps:

- principles,
- drivers,
- practices

We see here that he adopts something similar to what happened with the ten commandments. He starts laying out the fundamental principles that developers have to follow when moving to and applying XP, what we called before "values." The principles are simplicity, communication, feedback, and courage. They have little to do with the actual activities that people perform every day! However, they are extremely useful to define the overall mindset that people must have to

perform effectively such activities. This is quite the same as what happened to the people of Israel. The text «And the Lord spoke all these words: 'I am the Lord your God, who led you away from the land of Egypt, out of the house of servitude.'» does not specify any specific action to perform, but set the stage for the followup prescriptions. Moreover, quite like in the ten commandments, Kent Beck in his book anchors the principles to the experience of people, thus providing them substance.

Beck himself insists that these values are not a decorative preamble that one reads once and then forgets while doing the "real" work: communication, simplicity, feedback, courage, and, in the later editions of the book, respect, are supposed to stay in the background against which every decision on the project gets checked [4]. Pair programming is adopted because it maximises communication and feedback between two minds looking at the same problem. Design is kept as simple as possible at every step because complexity that is not yet needed is a debt against simplicity. The team commits to daily, even hourly, integration of the code because waiting to find out whether things work together betrays feedback. None of this is arbitrary and none of it is justified practice by practice: exactly as, at Sinai, the commandments are not justified precept by precept, but are all anchored first to the single value of a God who has already led the people out of the house of servitude.

After the principles, Kent Beck presents his drivers: focus on value, constant flow of activities, and no defects. The drivers resemble closely the fundamental directions we need – they are the bridge, exactly as the ban on strange gods or the ban on taking the Lord's name in vain was the bridge for the people of Israel. They are not yet concrete gestures that a developer performs at the keyboard, but they are already less abstract than a bare value such as "courage." The driver "focus on value," for instance, translates the value of communication into the disposition to keep talking, continually, about what actually matters to the customer rather than about what is convenient to build. The driver "no defects" translates the value of feedback into the disposition to treat every discovered error as information to act on immediately, rather than something to accumulate and rationalise away.

Finally, and only finally, there is a list of concrete practices: the planning game, small releases, metaphor, simple design, testing (writing the test before the code), refactoring, pair programming, collective ownership, continuous integration, a sustainable pace, an on-site customer. Take any of these practices alone, stripped of the values and drivers behind it, and it looks arbitrary, even bizarre – why should two people sit at one keyboard? why should the customer be physically present at all times? – exactly as a commandment against boiling a kid in its mother's milk looks arbitrary to a reader of Exodus who has not been given the values it serves. Only because the practice is read back through the driver, and the driver

back through the value, does it stop being an idol – a rule kept because it is a rule – and become instead the visible, workable expression of something the team has actually agreed to care about. Here too there is a written agreement: the practices of XP are set down, discussed, and adopted explicitly by the team, in the same spirit in which the tables of the law were a written, not just spoken, commitment.

This is why Beck's ordering matters more than it looks at first sight. A team simply handed the practices of XP – write tests first, pair, integrate continuously, plan iteratively – without ever sharing the values that motivate them, is in exactly the position of the Israelites who received a golden calf instead of a covenant: they have something to do, a ritual to perform, but nothing to believe in, and the first time the practice becomes inconvenient under pressure there is nothing left to appeal to except the naked rule, which then collapses. A team that has actually internalised communication, simplicity, feedback, and courage can instead improvise a new practice, drop an old one, or adapt the letter of the method to a situation Beck never anticipated, and still remain unmistakably an XP team – because what made it so was never the letter of the practice but the value the practice was written to serve.



Figure 3.4: A software team tracing the same cascade – from a shared value down to a concrete practice – that the tables of the law once carried from Sinai to the camp, after Kent Beck's structuring of XP [4].

3.2.4 The quest for hysteric rules

We have seen that effective changes require a deep and involved process, that encompasses sharing overarching values, identifying fundamental directions, and only later defining set of practices conflicts with the natural desire of people for "immediate" rules that can fill them with (very often very) short term emotions. We can see it very well analysing what happens to the Hebrew when Moses goes to meet God for a little while.

Original text [20]	**English translation [39]**
Videns autem populus quod moram faceret descendendi de monte Moyses, congregatus ad Aaron dixit: «Surge, fac nobis deos, qui nos praecedant; Moysi enim, huic viro, qui nos eduxit de terra Aegypti, ignoramus quid acciderit».	After the people saw that Moses had been on the mountain for a long time, they went to Aaron and said: «Make us an image of a god who will lead and protect us. Moses brought us out of Egypt, but nobody knows what has happened to him.»
Dixitque ad eos Aaron: «Tollite inaures aureas de uxorum filiorumque et filiarum vestrarum auribus et afferte ad me». Fecitque omnis populus, quae iusserat, deferens inaures ad Aaron. Quas cum ille accepisset, formavit stilo imaginem et fecit ex eis vitulum conflatilem.	Aaron told them: «Bring me the gold earrings that your wives and sons and daughters are wearing.» Everybody took off their earrings and brought them to Aaron, then he melted them and made an idol in the shape of a young bull.
Dixeruntque: «Hi sunt dii tui, Israel, qui te eduxerunt de terra Aegypti!».	All the people said to one another: «This is the god who brought us out of Egypt!»
Quod cum vidisset Aaron, aedificavit altare coram eo et praeconis voce clamavit dicens: «Cras sollemnitas Domini est». Surgen tesque mane altero die obtulerunt holocausta et hostias pacificas; et sedit populus manducare et bibere et surrexerunt ludere.	When Aaron saw what was happening, he built an altar in front of the idol and said, «Tomorrow we will celebrate in honour of the Lord.» The people got up early the next morning and killed some animals to be used for sacrifices and others to be eaten. Then everyone ate and drank so much that they began to carry on like wild people.

Quote 3.7: **Exodus** 32, 1-6

In the middle of the change, some problem arise that requires time to fix. But people cannot wait. They start feeling unsure. They cannot reason, they need to be reassured, like children looking for their mother. They reach the point of negating the evidence that God via Moses saved them. They ask Aaron to make a god for them (in Latin is plural, gods), god(s) that can lead and protect them. This request is completely paradoxical: how can a man build gods that lead and



Figure 3.5: The Israelites dancing and feasting around the golden calf at the foot of the mountain.

protect them? The man building such gods would be himself a god. But they do not care. There is a clearly a literal hysteria emerging from the whole situation. Changes induce mass reactions that are hardly rational, but still refer to some alleged previous experience. The people were used to have human-based god(s), and any time something was going in the right way, they were attributing such positive situations to the gods, while when something was going wrong they were assuming that some other god(s) was/were upset and they were acting accordingly. Now, something in the "new way" is not going the way they wanted, so simply the "good old way" is the solution. As we discussed in Chapter 2, they are ready to sacrifice themselves for it.

It is now interesting to see the reaction of Aaron. There is no doubt that Aaron is on the side of Moses, it is a proposer of change. However, his reaction in the middle of the craziness is resolve the situation trying to accommodate the requests of people, or, at least, compromise with them. In the hope that they would placate themselves, he asks for expensive action, probably thinking that the natural egoism of people would prevail – he asks people to being their most valuable pieces of jewellery, and the ones of their family members. It is important to stress that he is not appealing to rationality, but to a personal instinct of self-saving. He does not understand that, as mentioned by Tolstoy, people are ready for heroic and useless (or, even, damaging) actions.

Needless to say, the situation is similar to the one in Section 2.3.2 in the commentary of Berlin about Tolstoy «... those who performed acts of incredible self-

sacrifice or heroism [...] were the most useless.[2]» (III [6]) In this case it is written: «The people got up early the next morning and killed some animals to be used for sacrifices and others to be eaten.» (Exodus 32, 1-6 [39]): the writer evidences clearly that while some animals were killed to be eaten, other for useless sacrifices, which means, for nothing, in an act of "incredible self-sacrifice or heroism" since the people were in the desert, with not much food, and, in fact, in the following of the narration they will also suffer for hunger.

Aaron tries further to accommodate the mass, but with no result, the mass gets wilder. Therefore, we learn that in the wisdom of this narration, change is not supported by compromising or by letting people express the hysteria. Moses will behave differently, as we will see soon.

3.2.5 Leadership vs. hysterism

God refers to Moses what is going on while he is away and starts to make various statements, as if he wants to prove the wisdom of Moses.

Original text [20]	**English translation [39]**
Locutus est autem Dominus ad Moysen: «Vade, descende; peccavit populus tuus, quem eduxisti de terra Aegypti. Recesserunt cito de via, quam praecepi eis, feceruntque sibi vitulum conflatilem et adoraverunt atque immolantes ei hostias dixerunt: 'Isti sunt dii tui, Israel, qui te eduxerunt de terra Aegypti!'».	The Lord said to Moses: «Hurry back down! Those people you led out of Egypt are acting like fools. They have already stopped obeying me and have made themselves an idol in the shape of a young bull. They have bowed down to it, offered sacrifices, and said that it is the god who brought them out of Egypt.»
Rursumque ait Dominus ad Moysen: «Cerno quod populus iste durae cervicis sit; dimitte me, ut irascatur furor meus contra eos et deleam eos faciamque te in gentem magnam».	«Moses, I have seen how stubborn these people are, and I'm angry enough to destroy them, so don't try to stop me. But I will make your descendants into a great nation.»

[2]War and Peace, vol. 4, part 1, chapter 4.

Original text [20]	**English translation [39]**
Moyses autem orabat Dominum Deum suum dicens: «Cur, Domine, irascitur furor tuus contra populum tuum, quem eduxisti de terra Aegypti in fortitudine magna et in manu robusta? Ne, quaeso, dicant Aegyptii: 'Callide eduxit eos, ut interficeret in montibus et deleret e terra'. Quiescat ira tua, et esto placabilis super nequitia populi tui. Recordare Abraham, Isaac et Israel servorum tuorum, quibus iurasti per temetipsum dicens: 'Multiplicabo semen vestrum sicut stellas caeli; et universam terram hanc, de qua locutus sum, dabo semini vestro, et possidebitis eam semper'».	Moses tried to get the Lord God to change his mind: «Our Lord, you used your mighty power to bring these people out of Egypt. Now don't become angry and destroy them. If you do, the Egyptians will say that you brought your people out here into the mountains just to get rid of them. Please don't be angry with your people. Don't destroy them! Remember the solemn promise you made to Abraham, Isaac, and Jacob. You promised that some day they would have as many descendants as there are stars in the sky and that you would give them land.»
Placatusque est Dominus, ne faceret malum, quod locutus fuerat adversus populum suum.	So even though the Lord had threatened to destroy the people, he changed his mind and let them live.
Et reversus est Moyses de monte portans duas tabulas testimonii in manu sua scriptas ex utraque parte et factas opere Dei; scriptura quoque Dei erat sculpta in tabulis.	Moses went back down the mountain with the two flat stones on which God had written all his laws with his own hand, and he had used both sides of the stones.

Quote 3.8: **Exodus** 32, 7-16

The situation is similar to when a company owner has decided, say, to move to Agile Methods. Typically, in the eyes of several company owners, such methods appears a way in which developers and software engineers gain more freedom and better working condition.[3] God is angry, it is like an angry CEO of a company who finds that people are heavily criticising a change s/he has enacted exactly because such change has created the freedom that allows them to criticise. It is like a parent saying to a child: "You misbehaved because you abused the freedom I gave you."

Moses does not disappoint God. He is able to oppose to the idea of God. The English translation here is not as strong as the Latin version; let's contemplate it: Moyses autem orabat Dominum Deum suum dicens. "Autem" means "but," is

[3]We have already discussed in other texts that indeed the adoption of Agile Methods triggers typically higher level of employee satisfaction. It might also give more apparent freedom. However, it is also true that the level of discipline and of commitment is significantly increased. However, here we are not discussing the reality, rather the perception of the owner



Figure 3.6: Moses interceding with God on the mountain on behalf of the people.

an adversative "but." It expresses an opposite view. And then "orabat Dominum Deum suum" means that he was praying the Lord his God. So Moses fully aware of being a human in front of God, still was asking to God to do something different. And this was done through rational argumentation. The wisdom of the people is therefore that in front to a reaction to a change, rather than punishment the leader should appeal to the reasoning, asking all key stakeholders, perhaps even the owner of the company, to stick to such reasonings.

This in an humongous step. A man opposes God, and is able to persuade God. And he does it not by flattery, not by pleading for pity, but by appealing to the rational side of God: he reminds God of the old promises, the covenant made to Abraham, Isaac, and Jacob, and asks, in effect, that God be consistent with himself.

The shape of the exchange as a whole matters too. It is as if God is provoking Moses, testing him, acting for a moment like a general demanding exemplary punishment for a lapse in discipline – "let me alone, that my wrath may wax hot against them, and that I may consume them." A lesser leader, faced with a superior threatening to raze the whole enterprise over a single lapse, has two options, and both are hysteric: join in the punishment, competing in righteous anger to prove his own loyalty, or cave in completely and disown the people to save himself. Moses does neither. He refuses the naked discipline offered to him, and recalls instead the principles: not the letter of any single commandment, but the substance of the promise the whole relationship rests on. He shows, in the middle of the worst crisis of the whole narrative, that lasting changes and lasting

communities are not sustained by exemplary punishment, however satisfying it might feel in the moment, but by long-term commitments that both parties hold themselves to remembering. And something does come of it: Moses does not return empty-handed or simply forgiven, he comes back down the mountain still carrying the written tablets, the law undiminished by the crisis it has just survived.

3.2.6 Hysteric rules and leadership in XP

The whole episode is really a contrast between two responses a leader can have to a crisis of confidence, and the contrast has a name: Moses versus Aaron.

When the people grow afraid and restless because their leader is late coming down from the mountain, Aaron's response is to give them something to do right away: melt the gold, cast the calf, build the altar, proclaim the feast. It is, in its own way, competent crisis management – fast, decisive, responsive to the mood of the room – and this is exactly what makes it dangerous, not just foolish. Aaron manufactures certainty where there is none, and he does it by producing an idol: something concrete, visible, and immediately worshippable, that spares people the harder and slower work of trusting a promise they cannot see. Moses's response, confronted in turn with an angry God ready to abandon the whole project, is the opposite: even at his angriest, even after smashing the tablets he has just carried down the mountain, he does not manufacture a substitute certainty. He goes back up the mountain and re-engages the relationship, the covenant, the reasoned appeal to a promise already made. Aaron resolves the anxiety of the moment; Moses refuses to resolve it falsely.

Most of us have watched some version of this happen inside a real team, so the software engineering parallel is not hard to draw. A project under sudden pressure – a missed release, an escalation from a client, key people leaving – creates the same anxious, leaderless vacuum that overtook the camp at the foot of Sinai, and it tends to produce the same two kinds of response. The Aaron response is the sudden mandate: an emergency process imposed overnight, a new tool bought and rolled out as the fix, a rigid checklist issued to reassure everyone, including the person issuing it, that something is under control. It is popular because it works immediately, in the sense that it gives frightened people something to do with their hands. But it fails for the same reason the golden calf fails: the mandate was never anchored to a value anyone had agreed to, only to the emotional need of the moment, and it will be worshipped only until the next crisis shows it, too, to be hollow. The Moses response is slower and less immediately satisfying, and to an anxious team it can look almost like inaction: it consists in holding the team, under pressure, to the values and principles it had already, calmly, committed to before the crisis began, and refusing to let panic invent a new golden calf of process just because the old certainties feel, for the moment, to have gone silent on the

mountain.



Figure 3.7: A modern golden calf, built overnight: a missed deadline, a hastily produced dashboard, and a team left no more reassured than before.

3.3 Setting the fundamental directions

We have used, throughout this chapter, the expression "fundamental directions" as the middle term of a triad – values, fundamental directions, practices – and this is probably the least intuitive of the three terms, so let us say plainly what it is. A value is an abstract commitment: courage, simplicity, the sanctity of a covenant, the dignity of a day of rest. A practice is a concrete, repeatable gesture: writing a test before the code, resting on the seventh day, not moving a neighbour's boundary stone. A fundamental direction is neither of these. It is the disposition that a value produces once it starts to be taken seriously in a particular domain: general enough to cover many situations, but specific enough to already rule some of them out. "Do not make for yourself an idol" is not yet a practice, it does not tell a single Israelite what to do on a given morning, but it is no longer the bare value of God's unicity either; it is that value already pointed, like a compass needle, at the concrete danger the community is going to face. In the same way, "focus on value" or "welcome change" are not yet the planning game or continuous integration, but they are the value of communication, or of courage, already aimed at the specific temptations a software team, rather than an ancient tribal confederation, is going to meet.

This is why fundamental directions matter more than their middle, almost administrative position in the triad suggests. Values alone are too abstract to guide a

decision under pressure, everybody agrees, in the abstract, that simplicity is good, and practices alone are too narrow to survive a situation their authors never anticipated. Fundamental directions let a value generate new, previously unwritten practices when the old ones no longer apply, and let an existing practice, when challenged, be defended or dropped by appeal to something more than its own inertia. A team that has really absorbed the fundamental direction of "no defects," for instance, does not need a rule for every new kind of defect a new technology introduces; it already knows, before any rule is written, roughly what it must do.

3.4 Exercise reasoning

We can now turn to a text that has nothing to do, on the surface, with methodology, but that deals, more directly than anything else in the Christian tradition, with the relationship between a value and its concrete embodiment we have been discussing.

Original text [20]	**English translation [39]**
In principio erat Verbum, et Verbum erat apud Deum, et Deus erat Verbum.	In the beginning was the Word, and the Word was with God, and the Word was God.
Et Verbum caro factum est, et habitavit in nobis: et vidimus gloriam eius, gloriam quasi unigeniti a Patre, plenum gratiae et veritatis.	And the Word became flesh, and he lived among us, and we saw his glory, glory like that of an only-begotten son from the Father, full of grace and truth.

Quote 3.9: **Gospel of John** 1, 1 and 1, 14

The Word, the *Logos*, is presented first in its most abstract, purely rational form: reasoned communication itself, prior to any particular utterance, "with God" and identified with God. In the vocabulary we have been using, it is a value in its purest state, unembodied, universal, not attached to any single gesture or circumstance. Then, in a single sentence, the abstraction is made flesh: the Word does not stay a proposition to be admired from a distance, it "lived among us," visibly, so that people "saw his glory." This is exactly the movement a value must undergo to do any real work in an organisation, whether it is the value of a covenant at Sinai or the value of communication inside a software team: sooner or later it must become flesh, exercised and embodied in a person's actual, observable behaviour, or it stays a word spoken into the air, admirable and inert. A principle stated in a company handbook but never practised by any manager anyone can see has not become flesh; it has stayed exactly where the golden calf's worshippers wanted their god to stay, at a safe and abstract distance, obliging nobody in particular.

The exercise this section's title refers to is precisely this: the disciplined, repeated, embodied living-out of a value until it becomes as recognisable, and as concrete, as a person one has actually seen walk among us.

3.5 Changes through empathic leadership

Everything discussed so far – the ten commandments, the golden calf, Moses's rational appeal to the covenant, Kent Beck's values and drivers – concerns leadership exercised from a certain height: God speaking from the mountain, Moses coming down with tablets, a methodology handed to a team. There is, however, an entirely different register of leadership, one that does not command from above but leads by placing itself, literally, below the people it leads. It is the clearest contrast there is to the hysteric, idol-manufacturing response we just examined in Aaron. This is the leadership Jesus exercises at the last supper, and it is empathic, not hysteric: it produces adherence not by fear, not by the sudden manufacture of an object to worship, but by service freely given and, as a result, love freely returned.

Original text [20]	**English translation [39]**
Ante diem autem festum paschae, sciens Iesus quia venit eius hora ut transeat ex hoc mundo ad Patrem, cum dilexisset suos qui erant in mundo in finem dilexit eos. Et cena facta, cum diabolus iam misisset in corde ut traderet eum Iudas Simonis Scariotis, sciens quia omnia dedit ei Pater in manus et quia a Deo exivit et ad Deum vadit, surgit a cena et ponit vestimenta sua et cum accepisset linteum praecinxit se, deinde mittit aquam in pelvem et coepit lavare pedes discipulorum et extergere linteo quo erat praecinctus.	Now before the feast of the Passover, Jesus, knowing that his hour had come to pass out of this world to the Father, having loved his own who were in the world, loved them to the end. And during supper, when the devil had already put it into the heart of Judas Iscariot, Simon's son, to betray him, Jesus, knowing that the Father had given all things into his hands, and that he had come from God and was going to God, rose from supper, laid aside his garments, and taking a towel, girded himself. Then he poured water into a basin and began to wash the disciples' feet, and to wipe them with the towel with which he was girded.

Quote 3.10: **Gospel of John** 13, 1-5

Showing empathy and compassion in this way is not giving up leadership, and it is not giving up the possibility of change. The gesture is, almost provocatively, the reversal of the ordinary posture of a master toward his followers: Jesus lays aside his garments as a servant would, takes on the lowest of household tasks,



Figure 3.8: Jesus washing the feet of a disciple at the Last Supper.

and performs it on every one of the men at the table, Judas included, knowing perfectly well what Judas is about to do. Compassion here does not mean approval or blindness: washing Judas's feet is not an endorsement of what Judas is going to do a few hours later. It means the deliberate choice to communicate through embodied service rather than through command, and to keep setting a tie with people even when one already knows they might break it. Empathic leadership of this kind does not mean accepting whatever is asked of the leader, any more than Moses's appeal to reason meant accepting whatever the frightened crowd at the foot of the mountain demanded. It means choosing a register of persuasion – example lived in the body, rather than an idol built for the eye – that produces adherence without producing fear.

Original text [20]	**English translation [39]**
«Si diligitis me, mandata mea servate.»	«If you love me, keep my commandments.»

Quote 3.11: **Gospel of John** 14, 15

The order of the sentence is not accidental, and it sums up the whole point of this chapter: love comes first, as its own condition, and only then, as its consequence, comes the keeping of the commandments. It is the exact inversion of the hysteric sequence around the golden calf, where a rule – build us a god, give us something to worship right now – is manufactured first, in the vain hope that emotional adherence, fear turned into ritual, will follow from it. Here the commandments are

kept because they are already loved, or, more precisely, because the one who gave them has already been experienced, concretely and bodily, as someone who loves. The practice (keeping the commandment) is again the last term of a sequence that begins with a value (love) and passes through a fundamental disposition (fidelity to the one who has already served you) before it ever reaches a specific, actionable rule. In software engineering terms, this is the difference between the leader who tries to get adherence to a process by mandate and the vague threat of consequences, and the leader who has already, so to speak, washed the team's feet: who has been seen doing the unglamorous, unrewarded work alongside everyone else, and has earned, through example rather than decree, a following that keeps the team's agreed practices not out of fear of being caught breaking them but out of a freely returned commitment to the person, and the values, behind them.

3.6 Some reflections

People like idols. Idols are comforting because they are immediate, visible, and ask only to be worshipped, not to be reasoned with – they do not demand the slow, patient work of understanding. But idols create hysteria, not long term change; they burn brightly for the length of a single crisis and leave nothing behind once it has passed. Traditional software development, when it goes wrong, often collapses into exactly this posture: we follow a rule blindly, simply because it is the rule, with no memory left of the value it was written to serve. It is tempting to explain this away by appeal to compensation, people follow the rule because they are paid to, and work for money has always been, for most of human history, a good enough reason to comply. But for knowledge workers this is not enough: no salary, by itself, has ever produced the kind of voluntary, self-correcting excellence a good software team shows under pressure. What produces it is the same thing that produced the endurance of the Jewish law across thousands of years, and the same thing that keeps a team faithful to its own agreed practices long after the manager who introduced them has moved on: a firm, long-term commitment that has actually been understood, mentally and emotionally, and not just imposed.

We have seen this commitment worked out twice in this chapter, in two different registers. Moses persuades the Lord not by threatening him or flattering him, but by remembering, out loud, a promise already made: a concrete, rational act of recovering a previous statement, rather than an appeal to raw authority. In doing so he shows that even God is willing to be subject to the dialectics of Moses, to be bound, in his own dealings with his people, by rationality of behaviour rather than by the arbitrary exercise of power. Love, as the Gospel of John presents it, works by a similar mechanism but from the opposite direction: not memory of a past promise, but an experience so immediate and embodied – a leader kneeling

to wash one's feet – that keeping the commandments which follow from it no longer needs any external enforcement. Robert Baden-Powell, writing for a much more modest purpose, in *Aids to Scoutmastership*, had already noticed a version of the same mechanism at the level of an ordinary boy and an ordinary adult leader: what moves a person to follow, durably and willingly, is not the giving of orders but the sense of being cared for by the person giving them; motivating people well, in every register we have examined, turns out to matter more than commanding them well.[4] This, in the end, is also where Kent Beck's ordering – values, then fundamental directions, then practices – rejoins the biblical and evangelical material discussed here: in every one of these cases, durable change comes not from starting with the practice and hoping the commitment follows, but from starting with the commitment and trusting the practice to follow from it.

3.7 The context

Before turning to the last two illustrations of this chapter, let me be honest about why they were chosen. Baden-Powell's Scout movement and the Belyaev fox experiment discussed below belong to worlds very different from Sinai, from the upper room, and from a modern software team, and I am not suggesting they are interchangeable expressions of one timeless law of human motivation. What they share, and what makes them useful here, is more modest: each is a documented, observable case where a durable change of behaviour was achieved, or failed, depending on whether it was rooted first in something the person or population being changed actually cared about, rather than in a rule imposed from the outside. Scouting is chosen because it is one of the most self-conscious, explicitly theorised attempts in the modern era to apply exactly this insight to the education of the young. The fox experiment is chosen because it shows, at the furthest possible remove from any question of belief or reasoning, that the same mechanism – sustained selective pressure applied consistently over time – can reshape a disposition all the way down to physiology, which gives some confidence that what we have been describing in Scripture and in software teams is a real, general phenomenon and not just a rhetorical trope.

3.8 Human virtue

Baden-Powell built the whole discipline of Scoutmastership on rejecting command as the primary tool of leadership over the young. A Scoutmaster, in his view, does

[4] See [14], discussed at greater length below.

not get good behaviour chiefly by ordering it and punishing its absence; he gets it by appealing to something already in the boy: his sense of honour, his desire to be seen as worthy of trust, his attachment to people he admires. In short, by appealing to virtue rather than fear.

«When a boy finds somebody who takes an interest in him he responds and follows where he is led, and it is here that hero-worship comes in as a great force for helping the Scoutmaster.» (Baden Powell, [14])

The mechanism Baden-Powell describes is, at bottom, the one we have been tracing since the beginning of this chapter: durable adherence is rooted in a relationship and in an appeal to the person's own values – honour, the desire to be worthy of someone's interest and trust – rather than in the bare issuing of orders. A Scoutmaster who commanded obedience the way Aaron manufactured the golden calf, giving the boys rules to placate their restlessness rather than a cause to believe in, would produce exactly what Aaron produced: compliance for as long as the threat or the novelty lasted, and nothing once it wore off. A Scoutmaster, or a technical lead, who instead earns hero-worship in the modest, non-idolatrous sense Baden-Powell intends – admired for competence and care rather than feared for authority – obtains something closer to what Moses obtained from a people he never stopped treating, even at his angriest, as reasoning partners in a shared covenant.

3.9 The Belyaev Experiment, that is, how to promote change

Everything argued so far in this chapter concerns change in human beings, who can, at least in principle, be reasoned with, appealed to, and persuaded. A natural question is whether the same mechanism – sustained, consistent pressure in the service of a clear and narrow criterion, rather than a rule imposed and enforced by fiat – can reshape something as apparently fixed as an organism's own nature, and not just its behaviour for the moment. The Soviet geneticist Dmitri Belyaev's decades-long experiment in domesticating the silver fox answers this about as unambiguously as an experiment can: selecting, generation after generation, purely for tameness toward a human handler reshaped not only the foxes' behaviour but, as an unplanned side effect, their anatomy and physiology too, producing in a few decades a change that free-living evolution would have taken millennia to reach. I develop this case in full in Chapter 5, *Integrating teams within organizations*, when discussing what it takes to turn a group of people who simply share a contract into a real team; here it is enough to note what it establishes on its own: that a real and lasting change, even one that looks at first as though it would require altering

something as fixed as an organism's nature, is possible, provided it is pursued consistently, over a long, sustained period, in the service of a single, clearly held value rather than a rule invented to placate an immediate anxiety.

3.10 Where are the goals?

It is fitting to end this chapter with the question its title poses throughout: once we set aside the comfort of the golden calf, once we refuse to let a rule stand in for the value it was meant to serve, where are the goals we are actually supposed to work toward?

The answer this chapter has argued for, in every one of its examples, is that the goals are wherever the shared values actually point, and nowhere else – certainly not wherever an imposed rule, however confidently proclaimed, happens to claim they are. Aaron's golden calf pointed nowhere at all beyond the immediate relief of the anxious crowd that demanded it; it was, literally, made of the community's own melted wealth and pointed back only at itself. Moses's tablets, by contrast, pointed at a covenant already tested by lived experience, a God who had already led the people out of the house of servitude, and that is why they could survive being smashed and rewritten and still mean the same thing. Kent Beck's practices point at values – communication, simplicity, feedback, courage, respect – that a team can restate for itself in a different project, a different language, a different decade, and still recognise as the same commitment. And the empathic leader who washes feet before ever asking for obedience points, deliberately, not at himself as an idol to be worshipped, but at the love that makes the commandment worth keeping in the first place.

So the goals of any real change – in a religious community, in a Scout troop, on a fox farm, in a software team – are never found by reading the rulebook harder, or enforcing the existing practice more strictly, or manufacturing a new one overnight to fill an anxious silence. They are found by going back, patiently and rationally, the way Moses went back up the mountain, to the values the community actually holds, and asking honestly what those values require now. A team, or a person, that keeps asking this question has not lost its way just because the path forward is unclear for a while. A team that has stopped asking it, and has settled instead for the comfort of an idol, has lost its way even when, for the moment, everything looks like it is running smoothly.

4
Leadership in software

In which two crowns go to war over who gets to give orders no one may refuse.

(Captain Lucian)



A storm flings the ship into a war between two crowns.



Moon king: "The star is mine by right!" Sun king: "Right? You've never held it!"



Captain, aside: "Neither has. An order is safe when it can't be refused."



The captain leads his crew quietly away from the noise.

Figure 4.1: Two crowns go to war over who gets to give orders no one may refuse.

Every book on software management I have ever opened, including some I have written myself, sooner or later starts talking about leadership as though the word

explained itself: a leader is someone who leads, a manager is someone whose orders get followed, and the whole apparatus of titles, org charts, and reporting lines exists to make sure the right person's orders are the ones that get obeyed. That assumption does not survive even mild scrutiny, and the management literature has too much of a professional stake in it to examine it honestly, so I am not going to look for the scrutiny there. I will look for it in a children's book instead.

Antoine de Saint-Exupéry's *Le Petit Prince* is, among other things, a small treatise on the varieties of adult self-deception, told through a child who visits, one after another, a set of tiny planets each inhabited by exactly one grown-up, each grown-up embodying a single way in which adults fool themselves about what they are doing with their lives. The first planet the little prince visits is ruled by a king, and this encounter, more than any other in the book, belongs at the center of a chapter on leadership in software, because the king is not a tyrant, not a fool, and not even, by his own lights, unreasonable. He is something more useful than any of that: a portrait of authority hollowed out from the inside while keeping, right up to the moment it is tested, every outward sign of being the real thing. I give the encounter below in full, first in Saint-Exupéry's own French and then in Katherine Woods's English translation [35, 36], because the argument I build afterward depends on details – the exact order in which the king issues his commands, the exact moment he checks his almanac – that a summary would flatten out, and that deserve to be read in the author's own words first.

Original text [35]	**English translation** [36]
Le roi siégeait, habillé de pourpre et d'hermine, sur un trône très simple et cependant majestueux.	Clad in royal purple and ermine, [the king] was seated upon a throne which was at the same time both simple and majestic.
– Ah! Voilà un sujet, s'écria le roi quand il aperçut le petit prince.	"Ah! Here is a subject," exclaimed the king, when he saw the little prince coming.
Et le petit prince se demanda: «Comment peut-il me reconnaître puisqu'il ne m'a encore jamais vu!» Il ne savait pas que, pour les rois, le monde est très simplifié. Tous les hommes sont des sujets.	And the little prince asked himself: "How could he recognize me when he had never seen me before?" He did not know how the world is simplified for kings. To them, all men are subjects.

Quote 4.1: **The Little Prince**, chapter X

4.1 An authority that can never fail

At a quick read the king's planet looks like a small comic set piece, a bit of whimsy on the way to the book's more famous encounters with the fox and the rose. Read



Figure 4.2: The king consults his almanac before ordering the sunset, at the exact hour it was always going to arrive.



Figure 4.3: The king reaches after the little prince as he leaves, inventing a new title rather than ask whether his rule accomplished anything at all.

Original text [35]	**English translation** [36]
– Approche-toi que je te voie mieux, lui dit le roi qui était tout fier d'être roi pour quelqu'un. Le petit prince chercha des yeux où s'asseoir, mais la planète était toute encombrée par le magnifique manteau d'hermine. Il resta donc debout, et, comme il était fatigué, il bâilla. – Il est contraire à l'étiquette de bâiller en présence d'un roi, lui dit le monarque. Je te l'interdis. – Je ne peux pas m'en empêcher, répondit le petit prince tout confus. J'ai fait un long voyage et je n'ai pas dormi... – Alors, lui dit le roi, je t'ordonne de bâiller. Je n'ai vu per- sonne bâiller depuis des années. Les bâillements sont pour moi des curiosités. Allons! bâille encore. C'est un ordre. – Ça m'intimide... je ne peux plus... fit le petit prince tout rougissant. – Hum! Hum! répondit le roi. Alors je... je t'ordonne tan- tôt de bâiller et tantôt de... Il bredouillait un peu et paraissait vexé.	"Approach, so that I may see you better," said the king, who felt consumingly proud of being at last a king over somebody. The little prince looked everywhere to find a place to sit down; but the entire planet was crammed and obstructed by the king's magnificent ermine robe. So he remained standing upright, and, since he was tired, he yawned. "It is contrary to etiquette to yawn in the presence of a king," the monarch said to him. "I forbid you to do so." "I can't help it. I can't stop myself," replied the little prince, thoroughly embarrassed. "I have come on a long journey, and I have had no sleep . . ." "Ah, then," the king said. "I order you to yawn. It is years since I have seen anyone yawning. Yawns, to me, are objects of curiosity. Come, now! Yawn again! It is an order." "That frightens me . . . I cannot, any more . . ." murmured the little prince, now completely abashed. "Hum! Hum!" replied the king. "Then I- -I order you sometimes to yawn and sometimes to–" He sputtered a little, and seemed vexed.

Quote 4.2: **The Little Prince**, chapter X

slowly, with an eye trained by years of watching how authority actually behaves inside organizations, it is one of the most exact descriptions of illegitimate power I know of anywhere in literature, and precisely because the king never once does anything a management consultant would flag as a mistake. On the surface he is a model of restraint: he does not fly into a rage when the little prince yawns in his presence, he simply converts the yawn into an order; he does not insist, out of pique, that the sun set at once to please him, he consults his almanac and waits for seven-forty, the hour at which, as we already know from the rest of the book, the sun on this tiny planet was always going to set anyway, because a planet that small lets its one inhabitant have a sunset whenever he moves his chair a few steps. Every order the king issues in this encounter is one he has already, quietly, checked in advance that events will bear out. He commands the little prince to

Original text [35]	**English translation** [36]
Car le roi tenait essentiellement à ce que son autorité fût respectée. Il ne tolérait pas la désobéissance. C'était un mo- narque absolu. Mais, comme il était très bon, il donnait des ordres raisonnables. «Si j'ordonnais, disait-il couramment, si j'ordonnais à un général de se changer en oiseau de mer, et si le général n'obéissait pas, ce ne serait pas la faute du général. Ce serait ma faute.»	For what the king fundamentally insisted upon was that his authority should be re- spected. He tolerated no disobedience. He was an absolute monarch. But, because he was a very good man, he made his or- ders reasonable. "If I ordered a general," he would say, by way of example, "if I ordered a general to change himself into a sea bird, and if the general did not obey me, that would not be the fault of the general. It would be my fault."

Quote 4.3: **The Little Prince**, chapter X

yawn again only after the little prince has already said he cannot help yawning. He commands him to sit down just as the little prince is looking for somewhere to sit. He commands the sun to set at the hour it was going to set anyway, command or no command. He even commands the little prince to interrogate him a heartbeat before the little prince, on his own initiative, was about to do exactly that.

This is not restraint. It is a method, a disciplined one, for making sure authority is never actually tested. A command issued only once its outcome is already assured is not a command in any sense that matters; it is a description, dressed up in the grammar of an imperative, of something that was going to happen anyway. The king has discovered, whether he admits it to himself or not, that the sure way to be obeyed always is to order only what would have happened without the order. He even has a theory to dignify the practice: authority rests on reason, he tells the little prince, one must ask of each person only what that person can give, a general ordered to turn into a sea bird could hardly be blamed for failing, and the fault in that case would lie with the king who gave an unreasonable order, not with the general who could not obey it. Taken at face value this sounds like real political philosophy, and a fairly sophisticated one. Taken as a description of how the king actually operates, it is exactly backwards. He does not calibrate his orders to what is reasonable in the abstract; he calibrates them, case by case, to what he has already privately confirmed will happen whether he orders it or not. Reasonableness, in his mouth, is not a constraint that might sometimes cost him an order going unobeyed. It is the mechanism that keeps every one of his orders out of any position where it could fail.

Compare this with what would actually count as legitimate authority and the contrast is stark. Legitimate direction – the kind a parent, a teacher, a command-

Original text [35]	**English translation** [36]
– Puis-je m'asseoir? s'enquit timidement le petit prince. – Je t'ordonne de t'asseoir, lui répondit le roi, qui ramena majestueusement un pan de son manteau d'hermine. Mais le petit prince s'étonnait. La planète était minuscule. Sur quoi le roi pouvait-il bien régner? – Sire, lui dit-il... je vous demande pardon de vous interroger... – Je t'ordonne de m'interroger, se hâta de dire le roi. – Sire... sur quoi régnez-vous? – Sur tout, répondit le roi, avec une grande simplicité. – Sur tout? Le roi d'un geste discret désigna sa planète, les autres pla- nètes et les étoiles. – Sur tout ça? dit le petit prince. – Sur tout ça... répondit le roi. Car non seulement c'était un monarque absolu mais c'était un monarque universel.	"May I sit down?" came now a timid inquiry from the little prince. "I order you to do so," the king answered him, and majestically gathered in a fold of his ermine mantle. But the little prince was wondering . . . The planet was tiny. Over what could this king really rule? "Sire," he said to him, "I beg that you will excuse my asking you a question–" "I order you to ask me a question," the king hastened to assure him. "Sire–over what do you rule?" "Over everything," said the king, with magnificent simplicity. "Over everything?" The king made a gesture, which took in his planet, the other planets, and all the stars. "Over all that?" asked the little prince. "Over all that," the king answered. For his rule was not only absolute: it was also universal.

Quote 4.4: **The Little Prince**, chapter X

ing officer, or, as I get to shortly, a manager exercises – is direction that could, in principle, go unheeded. It rests on reasons the person receiving it is free to examine and, in real cases, to reject; on consent that has actually been sought and could have been withheld; on evidence or experience tested against the possibility of being wrong. Because it could fail, its actually succeeding means something: it tells you the reasons offered were good enough, or the trust extended was warranted, or the experience invoked was sound, this time, to move another free agent to act. The king's orders can never tell us any of this, because they are built, with great and unacknowledged care, never to be put where failure was ever a live possibility. An authority that cannot fail is not a strong authority. It is not an authority at all, because the whole point of the concept – that someone's say-so makes a difference to what happens – has quietly been removed from it, while the outward form, the pomp, the ermine, the vocabulary of command, has been left intact.

This is also, I think, the only honest answer to why the king's laws and orders exist at all, on a planet with nobody to govern but a single visiting child and, we later learn, one old rat heard scratching at night. They exist for exactly one purpose, and it is not government: they exist to manufacture, continuously, the

Original text [35]	**English translation** [36]
– Et les étoiles vous obéissent? – Bien sûr, lui dit le roi. Elles obéissent aussitôt. Je ne tolère pas l'indiscipline. Un tel pouvoir émerveilla le petit prince. S'il l'avait détenu lui-même, il aurait pu assister, non pas à quarante-quatre, mais à soixante-douze, ou même à cent, ou même à deux cents couchers de soleil dans la même journée, sans avoir jamais à tirer sa chaise! Et comme il se sentait un peu triste à cause du sou- venir de sa petite planète abandonnée, il s'enhardit à solliciter une grâce du roi: – Je voudrais voir un coucher de soleil... Faites-moi plaisir... Ordonnez au soleil de se coucher...	"And the stars obey you?" "Certainly they do," the king said. "They obey instantly. I do not permit insubordination." Such power was a thing for the little prince to marvel at. If he had been master of such complete authority, he would have been able to watch the sunset, not forty-four times in one day, but seventy-two, or even a hundred, or even two hundred times, without ever having to move his chair. And because he felt a bit sad as he remembered his little planet which he had forsaken, he plucked up his courage to ask the king a favor: "I should like to see a sunset . . . Do me that kindness . . . Order the sun to set . . ."

Quote 4.5: **The Little Prince**, chapter X

experience of being obeyed, which the king then mistakes for the experience of actually ruling. This is why he is so transparently delighted at having, for the first time in the book's implied timeline, a subject to command at all – "tout fier d'être roi pour quelqu'un," as Saint-Exupéry puts it, consumingly proud of being at last a king over somebody – and why, when the little prince turns to leave, the king's first instinct is not to ask whether he has actually been a good king, or whether his rule has accomplished anything, but simply to invent a new title, minister of justice, ambassador, anything, that lets him keep doing the one thing his position actually consists of: issuing directives to somebody who will nominally receive them. This, in turn, is the real reason the little prince leaves, and his leaving should not be read as an act of youthful defiance against oppressive rule. Rebellion needs something substantial to rebel against, an order strong enough, or dangerous enough, or wrong enough, to be worth resisting. The little prince finds nothing of the kind. He looks for something to push against and cannot locate it: a kingdom that turns out, on inspection, to contain nothing at all beyond the king's own person and a single rat he can sentence to death and pardon in the same breath, because it is the only rat there is; laws that turn out, on inspection, never to have been tested against a single instance of possible disobedience; an authority that, the one time the little prince actually invites the king to exercise it – by pointing out, quite politely, that the king could order him to leave within the minute and that conditions seem entirely favorable – has nothing to say, because the king, faced for



Figure 4.4: The king on his tiny, empty planet, giving orders he has already made sure cannot fail.

once with a command whose outcome he cannot privately guarantee, simply falls silent. There is nothing here to defy. There is only, once you look closely, nothing.

4.2 The manager who never risks being wrong

I have spent this much space on a page and a half of a children's book because the king's method – issue only the orders you have already privately confirmed will be obeyed, and dress the whole thing up in the vocabulary of reason – is not a literary curiosity. It is a fully operational management technique, in daily use in software organizations that would be mortified to see it described this plainly. I have watched it, and most readers with any real time in the industry have probably watched it too, in at least three common disguises.

The first is the manager who ratifies. This is the person who waits, before giving any direction to a team, until it is already unmistakably clear which way the team was going to go on its own, and then announces that exact direction as though it had originated with her. The team, mid-sprint, has already converged informally on a particular technical approach, or a priority order for the backlog, or a way of splitting a difficult story; the manager, sensing which way the wind has settled, announces, in a meeting or a message with her name on it, precisely that approach as a decision. Nothing she says is false, and nothing in it can ever be shown wrong, for the same reason nothing the king says can be shown wrong: the direction was

never tested against the possibility of the team going some other way, because it was issued only once that possibility had already closed. Such a manager can compile, over a long career, an unbroken record of decisions that turned out right, and the record is, in its own hollow way, entirely real – and entirely worthless as evidence of judgment, because it was bought by never once pointing anyone toward anything that had a real chance of not happening anyway.

The second disguise is the manager who is careful to phrase every directive so that it can never actually be tested, challenged, or refused: a preference floated as a passing remark rather than a decision, a strong recommendation that stops just short of being an instruction. If the team follows it she can claim credit for having led; if the team quietly ignores it she can claim, just as sincerely, never to have really insisted. This is the king's trick of ordering the sun to set at the hour it was already going to set, moved into the standup meeting: authority kept safe by never once putting it at risk.

The third, and in some ways the most sympathetic, is the manager who has simply never been tested, because the team has never yet hit a moment where his direction and its own instincts genuinely pulled apart. He does not know, and cannot know, whether the confident authority he projects would survive an actual disagreement, any more than the king's authority had ever, before the little prince arrived, faced a subject who actually wanted something other than what the king was about to command anyway.

All three share the same structure as the king: authority kept safe by making sure, case by case, that it is never placed where it could fail, and mistaken – by the manager herself, and often by the people around her – for authority that was earned. It is worth being precise about what earned authority actually looks like in a software team, because it is not the absence of the behaviors above but their exact opposite. A tech lead earns real authority by being demonstrably right often enough, on questions that mattered, where being wrong would have cost something, that a team follows her not because her title demands it but because her track record has, when it counted, warranted the trust. Such a lead gives direction the team really could refuse: she proposes an architecture the team is free to argue against, insists on paying down technical debt the team could, in principle, simply deprioritize, calls a risky decision under real uncertainty and owns the outcome if it goes wrong. It is precisely because the direction could have gone unheeded, and because she visibly bears the cost when her calls turn out badly, that the team's continuing to follow her means anything at all. This, I think, is the entire difference between a manager and a leader that this chapter's title promises: not seniority, not charisma, not even technical skill on its own, but the willingness to give direction that can actually fail, and the discipline to have failed often enough, honestly enough, and instructively enough, to have earned the

trust that makes the direction worth following the next time.

4.3 Self-organization, obedience, and what remains open

It should be clear by now why I do not think the choice this chapter has been circling – self-organization, a team governing itself and choosing its own path, against obedience, a team executing direction handed down by an authority above it – can simply be settled by declaring, as software's own folklore likes to declare, that self-organization always wins. The king's planet is not an argument against obedience as such. It is an argument against a particular counterfeit of authority that dresses itself up as legitimate command while carefully avoiding ever being tested as one. A team that trades the king's hollow orders for a self-organizing structure has traded up, in that specific case, because there was nothing of substance underneath the king's authority to lose. But it does not follow – and I do not think Saint-Exupéry's fable supports the inference – that self-organization is therefore unconditionally better than every form of directed authority, including the earned kind described above, in every circumstance where a group of people has to decide how to act together. Whether self-organization is simply the better alternative to obedience-based order, or whether it depends on something else, something that has to be built up over time before a team can be safely trusted to govern itself, is a question this chapter has deliberately left open. I take it up directly in the chapter that follows, where a very different group of people, stranded far from home with no king, hollow or otherwise, left to command them, has to discover the hard way what that something else turns out to be.

Original text [35]	**English translation** [36]
– Si j'ordonnais à un général de voler d'une fleur à l'autre à la façon d'un papillon, ou d'écrire une tragédie, ou de se changer en oiseau de mer, et si le général n'exécutait pas l'ordre reçu, qui, de lui ou de moi, serait dans son tort? – Ce serait vous, dit fermement le petit prince.	"If I ordered a general to fly from one flower to another like a butterfly, or to write a tragic drama, or to change himself into a sea bird, and if the general did not carry out the order that he had received, which one of us would be in the wrong?" the king demanded. "The general, or myself?" "You," said the little prince firmly.
– Exact. Il faut exiger de chacun ce que chacun peut donner, reprit le roi. L'autorité repose d'abord sur la raison. Si tu ordonnes à ton peuple d'aller se jeter à la mer, il fera la révolution. J'ai le droit d'exiger l'obéissance parce que mes ordres sont raisonnables.	"Exactly. One must require from each one the duty which each one can perform," the king went on. "Accepted authority rests first of all on reason. If you ordered your people to go and throw themselves into the sea, they would rise up in revolution. I have the right to require obedience because my orders are reasonable."
– Alors mon coucher de soleil? rappela le petit prince qui jamais n'oubliait une question une fois qu'il l'avait posée.	"Then my sunset?" the little prince reminded him: for he never forgot a question once he had asked it.
–Ton coucher de soleil, tu l'auras. Je l'exigerai. Mais j'attendrai, dans ma science du gouvernement, que les conditions soient favorables. – Quand ça sera-t-il? s'informa le petit prince. – Hem! hem! lui répondit le roi, qui consulta d'abord un gros calendrier, hem! hem! ce sera, vers... vers... ce sera ce soir vers sept heures quarante! Et tu verras comme je suis bien obéi.	"You shall have your sunset. I shall command it. But, according to my science of government, I shall wait until conditions are favorable." "When will that be?" inquired the little prince. "Hum! Hum!" replied the king; and before saying anything else he consulted a bulky almanac. "Hum! Hum! That will be about–about–that will be this evening about twenty minutes to eight. And you will see how well I am obeyed!"

Quote 4.6: **The Little Prince**, chapter X

Original text [35]	**English translation** [36]
Le petit prince bâilla. Il regrettait son coucher de soleil manqué. Et puis il s'ennuyait déjà un peu : – Je n'ai plus rien à faire ici, dit-il au roi. Je vais repartir! – Ne pars pas, répondit le roi qui était si fier d'avoir un su- jet. Ne pars pas, je te fais ministre! – Ministre de quoi? – De... de la justice! – Mais il n'y a personne à juger! – On ne sait pas, lui dit le roi. Je n'ai pas fait encore le tour de mon royaume. Je suis très vieux, je n'ai pas de place pour un carrosse, et ça me fatigue de marcher. – Oh! Mais j'ai déjà vu, dit le petit prince qui se pencha pour jeter encore un coup d'$\frac{1}{2}$il sur l'autre côté de la planète. Il n'y a personne là-bas non plus... – Tu te jugeras donc toi-même, lui répondit le roi. C'est le plus difficile. Il est bien plus difficile de se juger soi-même que de juger autrui. Si tu réussis à bien te juger, c'est que tu es un véritable sage. – Moi, dit le petit prince, je puis me juger moi-même n'importe où. Je n'ai pas besoin d'habiter ici. – Hem! Hem! dit le roi, je crois bien que sur ma planète il y a quelque part un vieux rat. Je l'entends la nuit. Tu pourras juger ce vieux rat. Tu le condamneras à mort de temps en temps. Ainsi sa vie dépendra de ta justice. Mais tu le gracieras chaque fois pour l'économiser. Il n'y en a qu'un.	The little prince yawned. He was regretting his lost sunset. And then, too, he was already beginning to be a little bored. "I have nothing more to do here," he said to the king. "So I shall set out on my way again." "Do not go," said the king, who was very proud of having a subject. "Do not go. I will make you a Minister!" "Minister of what?" "Minster of–of Justice!" "But there is nobody here to judge!" "We do not know that," the king said to him. "I have not yet made a complete tour of my kingdom. I am very old. There is no room here for a carriage. And it tires me to walk." "Oh, but I have looked already!" said the little prince, turning around to give one more glance to the other side of the planet. On that side, as on this, there was nobody at all . . ." "Then you shall judge yourself," the king answered. "that is the most difficult thing of all. It is much more difficult to judge oneself than to judge others. If you succeed in judging yourself rightly, then you are indeed a man of true wisdom." "Yes," said the little prince, "but I can judge myself anywhere. I do not need to live on this planet." "Hum! Hum!" said the king. "I have good reason to believe that somewhere on my planet there is an old rat. I hear him at night. You can judge this old rat. From time to time you will condemn him to death. Thus his life will depend on your justice. But you will pardon him on each occasion; for he must be treated thriftily. He is the only one we have."

Quote 4.7: **The Little Prince**, chapter X

Original text [35]	**English translation** [36]
– Moi, répondit le petit prince, je n'aime pas condamner à mort, et je crois bien que je m'en vais. – Non, dit le roi. Mais le petit prince, ayant achevé ses préparatifs, ne voulut point peiner le vieux monarque : – Si Votre Majesté désirait être obéie ponctuellement, elle pourrait me donner un ordre raisonnable. Elle pourrait m'ordonner, par exemple, de partir avant une minute. Il me semble que les conditions sont favorables... Le roi n'ayant rien répondu, le petit prince hésita d'abord, puis, avec un soupir, prit le départ. – Je te fais mon ambassadeur, se hâta alors de crier le roi. Il avait un grand air d'autorité. «Les grandes personnes sont bien étranges», se dit le petit prince, en lui-même, durant son voyage.	"I," replied the little prince, "do not like to condemn anyone to death. And now I think I will go on my way." "No," said the king. But the little prince, having now completed his preparations for departure, had no wish to grieve the old monarch. "If Your Majesty wishes to be promptly obeyed," he said, "he should be able to give me a reasonable order. He should be able, for example, to order me to be gone by the end of one minute. It seems to me that conditions are favorable . . ." As the king made no answer, the little prince hesitated a moment. Then, with a sigh, he took his leave. "I make you my Ambassador," the king called out, hastily. He had a magnificent air of authority. "The grown-ups are very strange," the little prince said to himself, as he continued on his journey.

Quote 4.8: **The Little Prince**, chapter X

5
Integrating teams within organizations

In which, trapped with total strangers, they learn this chapter's real lesson: a team survives only as one, or not at all.

(Captain Lucian)



A whale swallows the ship, crew and all.



Castaway: "You learn to survive together, or not at all."



Captain: "Then let's stop surviving. Let's get out – together."



Strangers an hour ago, they break into daylight as one.

Figure 5.1: Trapped with total strangers, they learn this chapter's real lesson: a team survives only as one, or not at all.

Every organization I have worked in, studied, or just watched from the outside has one small, apparently trivial problem: it is made of people, and people do

not arrive fitted to the box that the organizational chart has drawn for them. A company hires a person, but what actually walks in the door is a whole biography, a temperament, loyalties that predate the contract, and, more often than we like to admit, a role the person has already built somewhere else and now carries into the new job like a mask she cannot take off. So the task of an organization is not simply to assign roles; it is to take already-formed, already-masked individuals and fold them into something that behaves, at least for the duration of the project, like one coherent entity. This task is real. It is hard, it fails in instructive ways, and it can also succeed. But when it succeeds, it is not because someone imposed the right structure from above – it is because the people involved earned, the hard way, a cohesion that no organizational chart could have given them for free.

5.1 Six characters, one company

Original text [31]	**English translation** [32]
Senza badare al fantasma che portano con sé, in se stessi, vanno correndo, pieni di curiosità, dietro il fantasma altrui! E credono che sia una cosa diversa	Without paying any attention to the phantom that they themselves carry inside them, they run about, full of curiosity, chasing after somebody else's phantom! And they believe it to be something different.

Quote 5.1: **Tonight We Improvise**

This is Luigi Pirandello, in *Questa sera si recita a soggetto* (*Tonight We Improvise*), putting into the mouth of one of his characters the peculiar blindness of people who spend their lives chasing an image of themselves projected by somebody else, while staying strangers to the image they themselves carry and impose, in turn, on everyone around them. It fits this chapter well, because it names, in one sentence, the confusion between the mask one is given and the person one actually is. And it is not an isolated thought in Pirandello – it is closer to an obsession. A few years earlier, in what remains his single most celebrated play, he had already staged this same obsession at length. It is that play, not *Questa sera*, that gives this chapter its central image: an organization trying, and only partly succeeding, to turn a handful of separately formed people into a single working whole.

Sei personaggi in cerca d'autore (*Six Characters in Search of an Author*, 1921) opens on a bare stage, where a theatrical company is rehearsing a play under the direction of, simply, il Direttore – the Director. Into the middle of this rehearsal walk six people who are not actors and who claim not to belong to any existing

play: a Father, a Mother, a Stepdaughter, a Son, and two younger children. They are, they insist, *characters* – creatures conceived by an author who then abandoned them before giving them a play to inhabit, and who are now condemned to wander in search of someone willing to finish the work of putting them together into a coherent drama. Each of them arrives already fixed in a role, in Pirandello's own vocabulary already wearing a face that cannot be taken off: the Father is forever the man who wronged the Stepdaughter's mother; the Stepdaughter is forever the girl whom poverty and the Father's earlier conduct drove into a house of prostitution, where the two very nearly meet again under circumstances neither of them can forgive. Unlike the actors of the company, who can put on and take off a hundred different parts over a career, a Pirandellian character is condemned to a single, unchangeable identity, precisely because that identity was never allowed to finish developing into the full complexity of a real, lived life. This is, in one of the play's most quoted exchanges, exactly what the Father tries to explain to the bewildered Director, who wants nothing more than to send these six intruders away and get back to rehearsing his own, much less troublesome, script.

Original text [29]	**English translation** [30]
Ma è qui il male! Nel linguaggio. Noi tutti abbiamo dentro un mondo di cose; ciascuno il suo mondo di cose! E come possiamo intenderci, signore, se nelle parole ch'io dico metto il senso e il valore delle cose come sono dentro di me; mentre chi le ascolta, inevitabilmente le assume col senso e col valore che hanno per sé, del mondo com'egli l'ha dentro? Crediamo d'intenderci; non c'intendiamo mai!	But that is just where the trouble lies! In the words! We all have within us a whole world of things, each man of us his own special world. And how can we ever come to an understanding if I put in the words I utter the sense and value of things as I see them, while you who listen to me must inevitably translate them according to the conception of things each one of you has within himself. We think we understand each other, but we never really do!

Quote 5.2: **Six Characters in Search of an Author**, Act I

The whole comedy – and it is, in a very dark register, a comedy – of *Six Characters* lies in this: the Director, who is a competent and reasonably well-intentioned professional, wants to treat the six intruders as material for a play, that is, as characters who can be adapted, rewritten, softened, reconciled with one another, and slotted into acts and scenes according to the ordinary conventions of the stage. The six characters, in turn, resist him at every step, not out of malice but because they are, quite literally, unable to be anything other than what they already and irreversibly are. The Father insists on telling his side of the story before the Stepdaughter can tell hers; the Stepdaughter interrupts him to accuse him of lying, or at least of softening what really happened; the Mother weeps uncontrollably

every time the events of the past are recalled, unable either to explain herself or to stop grieving. The Director tries repeatedly to impose an order, a sequence, a conventional plot with a beginning, a middle, and an end, on material that keeps escaping the shape he is trying to give it, because each character brings into the rehearsal room a self-contained, already-hardened truth that refuses to be just one voice among the others in a coherently authored scene.



Figure 5.2: The six masked characters interrupt the rehearsal, to the startled reaction of the actors and the Director.

It does not take much imagination to see, in this image, a description of what actually happens inside a great many organizations, and a rather more honest one than the diagrams in a project charter. A company, a department, a cross-functional team assembled for a new initiative is not unlike the Director's rehearsal room. It gathers people who each arrive with their own already-formed «world of things,» in the Father's phrase: their own professional history, their own loyalties to a previous team or manager, their own private account of who is to blame for the last failed project, their own accumulated grievances. And it expects a manager, a Scrum Master, a project lead, cast in the role of Pirandello's Director, to fold all of this into a single coherent performance just by assigning roles, scheduling meetings, and drawing up a plan. It very rarely works this cleanly. Anyone who has sat in a retrospective where two colleagues re-litigate, for the fourth sprint in a row, an old disagreement that has nothing to do with the current backlog has watched, in miniature, the Father and the Stepdaughter interrupting each other on Pirandello's stage: two people, each convinced her own account is complete, unable to hear the other's account as anything but an affront to correct. The

organization, like the Director, would very much like this friction to resolve itself into a tidy scene. It generally does not – not by fiat, not by scheduling it away, not by writing a process document that says how disagreements are to be handled.

But it would be a mistake to conclude from Pirandello that people are permanently locked into their masks, that a Father is forever only the Father and a Stepdaughter forever only the Stepdaughter, and that no group of separately formed individuals can ever become something more integrated than the sum of its private grievances. Pirandello's characters are frozen precisely because they are fictions who never had the chance to keep on living, whose entire being was arrested at the moment of a single, unresolved catastrophe. Real people are not like this. Real people, and real animals for that matter, keep on living, keep accumulating experience, and, given the right pressures over the right span of time, can change what looks at first sight like a fixed and permanent nature. The clearest evidence for this has nothing to do with organizations, or with people at all, and everything to do with foxes.

5.2 Breeding cooperation: Belyaev's foxes

I already used this experiment once, briefly, in Chapter 3, *Enacting processes*, as one piece of evidence that a disposition can change durably and not just for a moment. It deserves the fuller treatment here, because it shows exactly what it takes to turn a group of people bound only by a contract into an actual team.

In 1959, the Soviet geneticist Dmitri Belyaev began, at an experimental farm near Novosibirsk, a project that would eventually run for more than four decades, well beyond his own death in 1985, and that remains one of the cleanest and most cited demonstrations in all of biology of how fast, and how thoroughly, a population's behavioural disposition can be reshaped by selection [44]. Belyaev's material was the silver fox, *Vulpes vulpes*, a species farmed across Russia for its pelt and, in its wild and farmed state alike, notoriously skittish, aggressive, and unapproachable by humans. His method was deliberately, almost brutally simple: in each generation, he and his collaborator Lyudmila Trut selected for breeding only those foxes that showed the least fear and the least aggression toward a human handler, discarding every other trait – coat colour, size, temperament in any other respect – as irrelevant to the selection criterion. Nothing else about the animals was engineered or bred for. The only thing that mattered, generation after generation, was tameness.

What happened next is the reason the experiment is now taught in genetics courses across the world. Within a mere six to ten generations, some of the foxes were behaving less like wary wild animals and more like dogs: wagging their tails at the approach of a familiar handler, seeking physical contact, whimpering to

attract attention rather than fleeing or biting. This alone would have been a remarkable enough result – confirmation that a broad and complex behavioural disposition, tameness toward human beings, could respond to a narrowly targeted selective pressure far faster than anyone had expected. But the more startling finding, and the one that made Belyaev's farm-fox experiment famous well beyond specialist genetics, is that the foxes did not change only in behaviour. Selecting purely for tameness produced, as an apparently unintended side effect, a whole cluster of physical changes that nobody had selected for directly: floppy, dog-like ears in animals whose wild relatives have permanently erect ones; tails that curl upward instead of hanging straight; coats that develop irregular white, piebald patches instead of the uniform silver of the wild type; skulls and faces that grow, in later generations, visibly shorter and more rounded, closer in proportion to a puppy's than to an adult wild fox's; and, at the physiological level, measurably altered levels of circulating stress hormones and of serotonin, the neurotransmitter most closely associated with calm, non-aggressive sociability [44]. Selecting for one behavioural trait had reshaped, in a matter of decades rather than millennia, an entire correlated suite of anatomical, hormonal, and behavioural characteristics – precisely the suite of traits, as it happens, that distinguishes every domesticated species from its wild ancestor, from the dog and the wolf to the pig and the boar.



Figure 5.3: A wary wild silver fox alongside a tame, tail-wagging fox approaching a handler's hand, after generations of selection for tameness.

This piece of evidence is useful precisely because it carries none of the sentimentality that discussions of teamwork usually carry. Nobody told the foxes to cooperate. Nobody explained to them the business case for tameness, held a workshop on trust, or asked them to internalize a set of shared values. What changed

their disposition, thoroughly and heritably, was sustained selective pressure, applied consistently, generation after generation, toward a single outcome. This is the strongest scientific rebuttal to the pessimistic reading of Pirandello I opened with: an already-formed nature is not a fixed and final thing, immune to the environment it is placed in. Given the right selective conditions over the right span of time, even something as apparently hard-wired as an animal's wariness of strangers can be reshaped into its opposite, and the reshaping does not stop at the behaviour that was targeted – it radiates outward and changes the whole organism. If tameness and cooperative affect can be bred into foxes in six generations, there is no reason why a group of separately formed human beings, thrown together in a demanding shared enterprise and subjected not to genetic selection but to the equally real pressure of shared experience, cannot be reshaped, within the span of a single working life, into something with a collective disposition quite different from the one each of them walked in with.

But Belyaev's foxes only tell me that such change is possible; they cannot tell me how it happens in a human being, since a fox does not reflect on its own transformation and a human being does. For that I need a novel that is about nothing else.

5.3 Resurrection: change through experience, not command

Lev Nikolayevich Tolstoy's last major novel, *Voskreseniye* (*Resurrection*, 1899), tells the story of Prince Dmitri Nekhlyudov, a wealthy and comfortably self-satisfied nobleman who finds himself, quite by accident, sitting on the jury at the trial of a prostitute accused of poisoning a client. The woman on trial is Katyusha Maslova, whom Nekhlyudov recognizes, to his own horror, as a young servant girl he seduced and abandoned a decade earlier, setting in motion the long chain of degradation that has brought her, step by step, to the dock.

Nothing external compels Nekhlyudov to do anything about this discovery. The court convicts Maslova, largely through a procedural error, of a murder she did not commit, and sentences her to hard labour in Siberia; Nekhlyudov could simply go home, disturbed but unharmed, and resume the comfortable life he had been living before the trial. Instead, over the length of the novel, he does the opposite: he pursues every legal appeal available to overturn the sentence, gives away much of his own land and fortune, and eventually follows Maslova's convoy across Russia toward Siberia, not because anyone has ordered him to, and not because any doctrine or institution demands it of him, but because the accumulated weight of what he has seen and been forced to reckon with makes his previous way of living



Figure 5.4: Nekhlyudov, seated among the jurors, recognizes the woman in the dock as the servant girl he once seduced and abandoned.

unbearable to him.

Across nearly five hundred pages, Tolstoy is charting a transformation that comes from the inside, worked slowly and often painfully through the sheer accumulation of lived experience honestly confronted. It is not a conversion imposed from outside. Nekhlyudov does not become a different man because a priest, a judge, or a superior officer instructs him to; every institution the novel puts in his path – the court, the prison administration, the church, high society itself – either fails him outright or actively obstructs the change taking place in him. What moves him is what he sees with his own eyes: the degradation of the prisons he visits on Maslova's behalf, the arbitrary cruelty of a bureaucracy that convicts the innocent as readily as the guilty, and, above all, the plain fact of Maslova's own suffering, which he cannot argue himself out of once he lets himself look at it directly. The title Tolstoy chose is not decoration. Nekhlyudov's transformation is presented, deliberately, as a kind of resurrection: a man who was, in every sense that matters, spiritually dead – comfortable, complacent, incurious about the consequences of his own past actions – is brought back to a fuller life, not by an external miracle, but by being made, through experience, to actually see what he had spent a decade not looking at.

The novel's closing gesture makes the mechanism explicit. In its final pages Nekhlyudov, alone, at last opens a copy of the Gospels that he has been carrying but not reading, and finds in it, read now with the changed eyes of a man who has been through what he has been through, an answer he had been unable to reach

through any amount of abstract reasoning or social convention.

Original text [43]	**English translation** [40]
С этой ночи для Нехлюдова началась совсем новая жизнь, не столько потому, что он вступил в новые условия жизни, но потому, что все, что случилось с ним с этого времени, получило для него совсем другое, чем прежде, значение.	From that night an entirely new life began for Nekhlyudov, not so much because he had entered into new conditions of life, but because everything that happened to him from that time onward had, for him, a completely different meaning from what it had before.

Quote 5.3: **Resurrection**, Conclusion

Notice what this sentence does and does not claim. It does not say that Nekhlyudov's external circumstances were transformed by a decree, a promotion, a change of employer or of law. His «new life» begins, Tolstoy insists, because the *meaning* he could give to his conditions changed, not the conditions themselves, and it changed because he had lived through enough and reflected honestly enough on it to see it differently. This is the precise complement to what Belyaev's foxes demonstrate. The foxes show that a disposition can be reshaped by sustained pressure even without reflection; Nekhlyudov shows what the same process looks like from the inside, in a being capable of narrating its own transformation to itself. Put the two together and I have a fairly complete account of how a collective or individual nature actually changes: not through an order issued and obeyed, not through a policy adopted and enforced, but through sustained, honestly confronted experience – exactly the raw material that a demanding shared enterprise, a hard project, a difficult campaign, a long march, supplies in abundance. The next case is exactly this mechanism at work not on a single nobleman but on an entire army.

5.4 The Ten Thousand: earning a team

In 401 BC, Cyrus the Younger, a Persian prince contesting the throne of his brother Artaxerxes II, hired a force of roughly ten thousand Greek mercenaries, along with a much larger contingent of native Asian troops, to march deep into the Persian heartland and settle the succession by force. Among the officers of this Greek contingent was a young Athenian named Xenophon, who had joined the expedition more or less as an adventurous bystander and who later wrote its history, the *Anabasis*, or *The March Up-Country*, one of the great firsthand narratives to survive from classical antiquity. Cyrus's army met Artaxerxes' forces at Cunaxa, near Babylon, and very nearly won; but Cyrus himself was killed in the fighting,

and with his death the entire purpose of the campaign evaporated in a single afternoon. The Greek mercenaries – hired soldiers, drawn from a dozen different city-states with no particular loyalty to one another beyond their shared trade – found themselves stranded some fifteen hundred miles from home, deep inside hostile territory, with no employer left to pay them, no treaty protecting them, and a Persian satrap, Tissaphernes, eager to be rid of them by whatever means proved convenient.

What Tissaphernes actually did, shortly afterward, made the situation still worse: under a pretence of negotiation, he lured the Greek generals, including the overall commander Clearchus, to a parley and had them treacherously seized and executed. An army that had already lost its employer now lost its entire command structure in a single act of betrayal, and every reason of realism suggested that it should, at that point, simply disintegrate into a leaderless, doomed mob, to be picked off piecemeal by a hostile empire on its own ground. This is exactly the moment at which, according to Xenophon's own account, the Ten Thousand instead did something that no order from any king or general had told them to do: gathered in an assembly of their own initiative, and elected new commanders from among their own surviving officers, Xenophon himself among them, to lead them home.

Original text [47]	**English translation** [46]
Ἐντεῦθεν δὲ συνελθόντες οἱ στρατηγοὶ τῶν ἀπολειπομένων στρατιωτῶν καὶ λοχαγοὶ ἐβουλεύοντο τί χρὴ ποιεῖν, καὶ εἵλαντο ἄλλους στρατηγοὺς ἀντὶ τῶν ἀπολωλότων.	«Thereupon the officers and captains of the surviving troops assembled together to take counsel as to what should be done, and they chose other generals in place of those who had perished.»

Quote 5.4: **Anabasis**, Book III

Xenophon does not describe this as an act of desperation but as a deliberative one: the surviving troops meet, discuss what should be done, and choose from among themselves the men they judge most fit to lead them, precisely because no external authority remained to appoint anyone for them. Over the months that follow, this improvised, elected leadership guides the Ten Thousand on a fighting march of roughly a thousand miles, north through hostile Persian and Kurdish territory, over the mountains of Armenia in winter, harassed continually by local forces who had every reason to want them dead. Along the way, the army governs itself through repeated assemblies, deposes and replaces commanders who lose the soldiers' confidence, debates strategy openly rather than simply receiving it from above, and behaves, in Xenophon's careful account, less like a mercenary rabble and more like a self-governing polis on the march. And then, after months



Figure 5.5: The surviving officers and soldiers, with no king or general left to appoint anyone for them, gather in assembly to choose their own new leaders.

of exactly this shared ordeal, comes the moment every reader of the *Anabasis* remembers best.

Original text [47]	**English translation** [46]
Ἐπεὶ δὲ οἱ πρῶτοι ἐγένοντο ἐπὶ τοῦ ὄρους καὶ κατεῖδον τὴν θάλατταν, κραυγὴ πολλὴ ἐγένετο. [...] ἐβόων· Θάλαττα, θάλαττα.	When the vanguard reached the summit of the mountain and caught sight of the sea, a great shout went up. [...] they were shouting, «The sea! The sea!»

Quote 5.5: **Anabasis**, Book IV

The shout ran back through the whole column, man by man, from the vanguard that first crested Mount Theches down to the rearguard still fighting off pursuers miles behind. It marks an army that no longer answered to any single master recognizing that it had, through its own sustained effort, brought itself home.

The Ten Thousand did not begin as a team. They began as exactly what Cyrus had hired: a loose collection of professional soldiers from different cities, bound together by a wage and nothing else, no more a coherent whole at the outset than Pirandello's six characters are a finished play when they first walk onto the Director's stage. What made them into something else – an army capable of governing itself, of electing and replacing its own leaders, of marching a thousand miles through hostile country without falling apart – was neither an order from above, since the only order-giving authority they had ever had was dead within days of



Figure 5.6: The vanguard of the Ten Thousand, at the crest of Mount Theches, shouts at the first sight of the sea.

the campaign's start, nor some pre-existing natural unity they happened to share. It was the shared ordeal itself. Months of common danger, common hardship, and common decision-making did to this collection of mercenaries something close to what Belyaev's selective breeding did to his foxes and what accumulated experience did to Tolstoy's Nekhlyudov: sustained pressure, honestly lived through rather than commanded by fiat, reshaped a group bound only by a contract into something with a different, more cohesive collective disposition.

Here, then, is the thesis of this chapter, stated plainly. Self-organization – a group governing itself, choosing its own leaders, deciding collectively rather than simply executing orders handed down from above – does outperform rigid, imposed command, but not unconditionally, and the Ten Thousand show exactly why. It worked for them because, by the time they had to rely on it, at Tissaphernes' betrayal and after, they had already been forged, in the earlier stages of the campaign, into a group with shared motivation and earned mutual trust: they had fought together at Cunaxa, marched together, faced the same betrayal together, and elected leaders whose competence they had already had occasion to observe. Self-organization was not a magic recipe that would have worked for any random group of ten thousand armed strangers dropped into hostile territory. It worked because the cohesion it depends on had already been earned the hard way – the same way Nekhlyudov's transformation had to be earned through experience rather than declared by decree, the same way Belyaev's tame foxes had to be bred across real generations rather than willed into existence in one.

Compare this to the king that Antoine de Saint-Exupéry's Little Prince meets on the first of the strange little planets he visits. This king rules «sur tout» and yet has, as I discussed in the previous chapter, absolutely nothing under his actual command beyond an old rat he can sentence to death and then pardon out of necessity, because it is the only rat he has. He demands obedience as an end in itself and dresses it up as reasonableness – he never gives an order he suspects will be disobeyed – but he never earns the following he claims by right. The moment the Little Prince has somewhere else to be, he simply leaves, and the king's whole apparatus of command turns out to have governed nobody at all. So is self-organization simply, unconditionally, the better alternative to the king's obedience-based order, or does it need something more to work? The Ten Thousand answer this precisely. Self-organization is indeed better, because unlike the king's brittle authority it survives the sudden loss of its own leadership and keeps functioning, as it did the moment Clearchus and his fellow generals were murdered. But it is better only where real cohesion and real shared motivation have already been earned, unlike on the king's empty planet. Where that cohesion is absent, self-organization does not rescue a group any more than obedience would: a leaderless assembly of people who have not been through anything together falls apart just as readily as an army whose only bond was a king's arbitrary decree. Command is not simply the fallback that saves the day when self-organization runs out – a rigid, imposed order fails on exactly the same ground, the absence of earned motivation, on which ungrounded self-organization fails. What actually works, in an army stranded in Persia or in a team assembled inside an organization, is neither obedience nor spontaneity as such. It is the accumulated, lived, often painful process – Belyaev's generations of selection, Nekhlyudov's honestly confronted experience, the Ten Thousand's shared danger on the road home – through which Pirandello's separately masked characters can, given enough time and enough shared adversity, actually become the single company the Director was hoping for from the very first rehearsal.

6
Healing software (processes)

In which a court teaches this chapter's real lesson: diagnose the evidence before you pass judgment, the way a good doctor examines before prescribing.

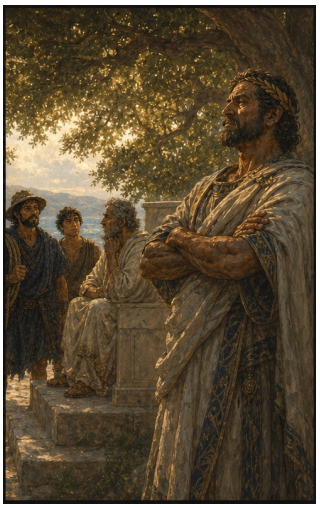
(Captain Lucian)



An island court, a judge, a case still open.



A witness speaks; the judge listens before deciding anything.



Dogmatist: "Why listen? I know the verdict." Judge: "Then you decided before you looked."



The captain's crew departs as the evidence, not the mood, wins.

Figure 6.1: The court that teaches this chapter's real lesson: diagnose the evidence before you pass judgment, the way a good doctor examines before prescribing.

The term "software engineering" describes our discipline accurately, because it

is indeed an engineering discipline: on the basis of some formal principles we build real systems that other human beings then rely upon. Let me emphasize the looseness of my statement, "some formal principles." The principles that ground an engineering discipline do not need to be correct in some absolute, final sense; they need to explain, in some replicable way, the phenomena they are meant to govern. Aristotelian physics is the clearest case. It held together the practical craft of building, moving, and predicting things for the better part of two thousand years, and it was wrong on almost every substantial point: there is no natural place toward which heavy bodies strive, motion does not require a mover in constant contact, the void is not in fact abhorred by nature. And yet it worked, because it let generations of engineers, physicians, and philosophers reason about the world in a shared, teachable, and reasonably predictive language. A principle earns its keep by being usable, not by being true: it lets a community of practitioners agree, disagree, correct one another, and hand down a discipline instead of a private intuition.

I insist on this because it undoes a common prejudice about engineering. Many people think of engineering, any engineering, as a rigid, top-down, waterfallish sort of endeavour, in which a plan is drawn once and for all and then simply executed. This is not true. I have already discussed at length, in the chapters that precede this one, that slipping schedules, errors in design, and failures in the running system are not a software-specific pathology; they happen in every engineering discipline the moment real complexity enters the picture, and bridges fall, planes are grounded, buildings sink, nuclear plants melt down. What distinguishes a mature engineering discipline from an immature one is not the absence of error, which is impossible, but a disciplined way of noticing, naming, and correcting error before it kills someone. This is precisely where software engineering has been unexpectedly poor, and an old sister discipline, considerably older than physics itself, has much to teach us here.

6.1 The missing word

We speak of software constantly in the vocabulary of the body and the mind – artificial minds, artificial intelligence, artificial life, systems we want to endow with something like consciousness, or at least with a convincing simulacrum of purposeful behaviour – and yet no one, as far as I know, has proposed the term **software medicine**. We are obsessed with replicating the human being in silicon; what could be closer to the human condition than getting sick, erring, needing care? And yet the medical metaphor, when it shows up in our field at all, is oddly narrow. The only place it has really naturalized itself in our vocabulary is the pairing of "virus" and "antivirus": a hostile, external agent of infection and the

specialized countermeasure built to destroy it. It is a vivid metaphor, and not a wrong one, but it is a thin slice of what medicine actually is. Medicine, for most of its history, and for most of what a physician actually does even today, is not mainly the science of killing invaders. It is the art of reading a body that is not behaving as expected and figuring out, often under great uncertainty, what is actually wrong with it before doing anything about it. This fuller sense of medicine, the one that precedes and grounds even the notion of a cure, is what I want to bring into software engineering here, because it maps onto our discipline far better than "virus" ever did.

6.2 Hippocrates and Galen: the fox and the hedgehog of medicine

I have already introduced, discussing Tolstoy through the eyes of Isaiah Berlin, the ancient distinction between the fox, who knows many things and adapts what she does to what she senses, and the hedgehog, who knows one big thing and moves according to it regardless of circumstance. The history of Western medicine offers maybe the cleanest example of the same distinction, this time among physicians rather than historians or philosophers, and playing out over the better part of two millennia rather than a single lifetime: Hippocrates against Galen.

Hippocrates, and the loosely federated body of texts collected under his name as the Hippocratic Corpus, practiced and taught a medicine of the fox. It is a medicine of watching. The physician sits at the bedside, records the course of a fever hour by hour, notes the colour of the skin, the smell of the breath, the character of the pulse, and only very cautiously moves from these particulars to a judgment about what to do. The Hippocratic physician does not know the one big thing that explains every illness; she knows many small things, and she has to weigh them against one another, aware that the same sign can mean different things in different bodies and that experience itself can mislead. This epistemic humility is stated so severely that it has become, deservedly, the most quoted sentence in the whole history of medicine, the opening of the *Aphorisms*:

Original text [17]	**English translation** [21]
Ὁ βίος βραχύς, ἡ δὲ τέχνη μακρή, ὁ δὲ καιρὸς ὀξύς, ἡ δὲ πεῖρα σφαλερή, ἡ δὲ κρίσις χαλεπή.	«Life is short, and Art long; the crisis fleeting; experience perilous, and decision difficult.»

Quote 6.1: **Aphorisms**, I, 1

Notice the word I have set in emphasis: experience is perilous, ἡ πεῖρα σφαλερή.



Figure 6.2: The Hippocratic physician at the bedside: attentive observation of the pulse and the patient's condition, before any judgment is made.

The Hippocratic physician does not trust the raw datum of the senses as if it spoke for itself; a sign is treacherous precisely because it must be interpreted, and the interpretation, the κρίσις, the judgment, is the hard and dangerous part. Nobody here has a master theory that dissolves every case into a known category. Hippocrates reads, case by case, the evidence the body offers, and accepts that the reading will sometimes go wrong.

Galen, roughly six centuries later, is the hedgehog. He inherited a great deal of Hippocratic vocabulary and even some Hippocratic humility in his best pages, but the system he built, and above all the system that later centuries extracted from him and canonized, goes in the opposite direction: a single, comprehensive, confidently teleological doctrine. Every organ has its assigned purpose. Every temperament reduces to a proportion among four humours – blood, phlegm, yellow bile, black bile. Every departure from health can, in principle, be explained and corrected by restoring the right balance among them. The doctrine is held together by a governing axiom that Galen repeats throughout the treatise *On the Usefulness of the Parts of the Body*, inherited in turn from Aristotle:

Original text [12]	**English translation** [25]
Ἡ φύσις οὐδὲν μάτην ποιεῖ.	«Nature does nothing in vain.»

Quote 6.2: **On the Usefulness of the Parts of the Body**

This is a beautiful sentence, and as a heuristic it is not even a bad one: looking for the function of a structure has driven real discoveries, in anatomy and later in evolutionary biology. But taken as an axiom rather than a heuristic, nature does nothing in vain becomes a machine for explaining away every counterexample. Whatever the body does, there must be a purpose for it, and whatever the theory of the four humours cannot immediately explain can always be patched with a further, ad hoc, purposeful just-so story. Galenic medicine, fortified this way, was taught essentially unchallenged in the universities of Europe, defended by the authority of the medical faculties and by fourteen centuries of institutional repetition, well into the seventeenth century – and it was almost entirely wrong about how the body actually works. Bloodletting, the signature therapy that followed logically from the humoral doctrine, probably killed more patients than it cured, including, if the anecdote is to be trusted, George Washington. It took a fox, William Harvey, who spent years watching, measuring, and dissecting, to show that the blood circulates instead of being consumed and regenerated according to appetite. Even then the Galenic edifice did not collapse quickly: an institutionalized hedgehog dies very slowly, because its confidence has been made structural rather than personal.



Figure 6.3: Galenic confidence in practice: bloodletting performed as a matter of doctrine, surrounded by anatomical charts.

6.3 Semeiotics: reading the signs before treating them

The word I want to rescue from this history is *semeiotic*, from the Greek σημεῖον, *sēmeion*, "sign." Semeiotics – sometimes spelled, in its older medical sense, *semiotics*, though I prefer the fuller spelling to keep it apart from the modern discipline that studies signs in language and culture – was, for the ancient physicians, the branch of the art devoted to reading the signs the body offers: the pulse, the colour of the skin, the pattern of a fever, the quality of the urine. From these signs, and before any treatment is attempted, the physician infers what is actually going on inside a patient she cannot open up to look at. This is diagnosis in its original sense: *dia-gignōskein*, to discern thoroughly, to tell things apart. Hippocrates is the semeiotician par excellence among the founders of Western medicine: his whole method is to become a careful, humble reader of symptoms and to resist the temptation to leap to a cure before the reading is done. Galen too wrote extensively on semeiotics, and on the pulse in particular, and it would be unfair to deny him a real diagnostic talent. What went wrong was not the practice of reading signs. It was the theory used to read them, which calcified into a system so complete that it stopped being surprised by its own patients. A diagnostic doctrine that has become diagnosis-proof, one that always has an explanation ready no matter what the signs actually say, is not semeiotics any more. It is a creed.

The same distinction holds, almost exactly, for how we, as software engineers, read the signs our own systems and our own teams give us. A code smell is a sign, not a disease. Duplicated code, a class that has grown improbably large, a module that nobody dares touch, are not the problem themselves; they are symptoms of something else, a missing abstraction, an organizational boundary drawn in the wrong place, a requirement nobody understood well enough to name. A recurring friction inside a team – the same argument resurfacing sprint after sprint, the same handoff that always goes badly, the same feature that is always late – is a sign in the Hippocratic sense: the process is trying to tell us something, if we are patient enough to read it instead of treating it away. The Galenic temptation in our own field is familiar. The moment we adopt a single doctrine, a specific flavour of Scrum, a specific religion of microservices, a specific dogma about test coverage, strongly enough that it becomes diagnosis-proof, we stop reading our own signs and start explaining them away. Every failure of the method then gets blamed on an insufficiently faithful application of the method, the same way every failure of bloodletting was blamed on imprecise bleeding rather than on the theory of the four humours itself. Software semeiotics would be the deliberate, humble, case-by-case cultivation of the Hippocratic habit: read the signs first, resist the ready-made explanation, and only then decide what, if anything, to do.

6.4 Cure and long-term care

This brings me to a distinction that medicine has always kept and that we, in software, tend to collapse: the distinction between a cure and long-term care. A cure is an acute intervention aimed at a single, identifiable episode: the antibiotic for the infection, the surgery for the appendix, and, in our own vocabulary, the hotfix for the crash in production, the emergency rewrite of a module that has become unmaintainable. A cure should be fast, targeted, and final. Long-term care is something else. It is the sustained, unglamorous, never quite finished practice of attending to a living system over time: managing a chronic condition instead of eliminating it once and for all, accepting that a codebase, like an ageing body, keeps generating new signs to be read for as long as it lives. The job of the engineer, like the job of the physician, does not end when the crisis ends. Continuous refactoring, retrospectives that are actually acted upon, the slow cultivation of a team's shared understanding of its own system, are all instances of long-term care, and they take just as much skill as the dramatic act of curing an acute failure, even if they are less glamorous. A discipline that only knows how to cure and never learns how to care will keep manufacturing the same crises, because it never developed the semeiotic patience to notice the small signs that precede them.

This is why I have called this chapter "Healing software (processes)" rather than simply "Fixing software." To heal is to attend to the whole course of an illness, from the earliest and most ambiguous signs to the slow work of recovery and maintenance, not just to silence a symptom. Most of the time it requires being a fox. You watch, you read the signs, you revise your judgment as new signs appear, and you resist the one big theory that would make every case identical to every other. Semeiotics comes before therapy, and long-term care goes alongside the cure, and sometimes replaces it. Aristotelian physics, with which I opened this chapter, survived for two thousand years because it kept being useful, not because it was true, and it gave way in the end to patient, Hippocratic observation of the phenomena it could no longer explain, not to a louder assertion of confidence. I suspect the same fate, and the same remedy, awaits every one of our own methodological orthodoxies. In the meantime the most useful thing we can do is keep a physician's eye on the software processes we build, and on the teams that inhabit them.

7
Back to the roots

In which every island turns out to have taught the same lesson, differently dressed.

(Captain Lucian)



The last island falls behind them, and the crew is still whole.



They look back: every island taught the same lesson, differently dressed.



Captain, to the reader: "What happened in the other world I shall tell you in the succeeding books." [38]



The ship sails into open mist. Nothing ahead is shown.

Figure 7.1: Every island taught the same lesson in a different costume — and the captain, like Lucian himself, leaves the rest of the story for you to write.

I opened this book with a myth, and I close it by naming the myth again before showing what replaces it. The myth is that we turn to lean and agile approaches because we are not good enough for something more rigorous: not good enough to plan properly, so we resort to the planning game; not good enough to specify properly, so we settle for user stories; not good enough to document properly, so we tell ourselves the code should speak for itself. I called this the not good enough myth in my introduction. Six chapters later, across epistemology, anthropology, children's literature, organizational drama, evolutionary biology and the history of medicine, the myth does not hold up, and here, once and for all, in the most exact words I can find, is what takes its place: lean/agile adoption isn't a maturity deficiency but a more effective way of managing uncertainty, proven by old wisdom, not driven by the fear-of-uncertainty constructs that only give an illusion of control and don't actually equip us with effective mechanisms to manage uncertainty. That is the thesis of the whole book, worked out chapter by chapter, and this final chapter only has to show it is one thesis and not six loosely related ones.



Figure 7.2: A single old tree, its visible roots already carved with the ship, the fox, and the tablets this book has been excavating all along.

Let me be precise about what I am and am not claiming. I am not claiming lean and agile methods are beyond criticism, or that a planning game, a user story, a self-organizing team automatically inherits the wisdom of Socrates or Hippocrates just by adopting the right vocabulary. Whole chapters of this book describe failure modes that are just as real inside agile teams as inside the traditional ones they replaced: the zealot who turns a useful practice into an unquestionable rule, the hysteric change that skips values and goes straight for a fabricated idol, the hollow

authority that issues only orders it already privately knew would succeed. My claim is narrower, and I think more durable: the reason to prefer adaptive, evidence-based, values-first ways of working is not that we failed to reach some higher, more rigorous alternative, but that the higher, more rigorous alternative was mistaking the appearance of control for the substance of it, all along. The old wisdom I went looking for in this book was never proposing agility as a consolation prize. It proposed it, under other names and in other centuries, as the thing that actually works when the future will not sit still.

7.1 A gallery of the same distinction

Let me walk back through the six chapters in order. I am not re-arguing any of them – each already spent its own space earning its own conclusion – I am just pointing, in a sentence or two each, at the deeper pattern each chapter's specific image belongs to. There are only three such patterns in this book, and every chapter belongs to one of them: a pattern about **knowing**, a pattern about **changing**, and a pattern about **leading**. Naming them here, chapter by chapter, is the job of a conclusion.

The introduction belongs to the pattern of knowing, though I did not put it that way when I wrote it. The not good enough myth is not just an unflattering description of agile teams. It is a false claim about what we need to know before we are entitled to act: that responsible action requires, as a precondition, a complete and formalizable grasp of the future, something no one ever actually has. Chapter 2 takes up the same pattern directly and gives it its clearest vocabulary. There is Socrates, on trial for admitting his own ignorance. There is Dante, who knows exactly where rational knowledge ends and love has to take over, and who lets Ulysses's fate (Dante's Ulysses, not Lucian's) show what happens when that boundary is crossed by narcissism rather than humility. And underneath both there is Archilochus's fox and hedgehog, as Isaiah Berlin recovers them from Tolstoy: the fox who knows many things and adapts, the hedgehog who knows one big thing and insists on it. Chapter 6 tells the same story again, this time as the history of Western medicine. Hippocrates read the fleeting, treacherous signs of a body he could not open, and refused to let doctrine make his diagnosis for him. Galen worked from the confident axiom that nature does nothing in vain, and that axiom survived fourteen centuries of institutional repetition while being wrong on nearly every substantial point. Hippocrates is the fox reading signs; Galen is the hedgehog, confident and mistaken. The distinction is the same one, only restaged in the clinic instead of the courtroom or the library.

Chapter 3 belongs to the pattern of changing, and it needed a different cast because the question is different: not what can we know, but how does a group

of people actually come to behave differently, and keep behaving that way. The Exodus narrative answers this in the order it insists on – values stated first and anchored in lived experience, fundamental directions derived from those values, and only at the end, specific practices. I noted there that Kent Beck lays out Extreme Programming in the same order. The same chapter also gives the counter-case: the Golden Calf, a hysteric, idol-driven attempt at instant change that skips values and principles entirely and jumps to a fabricated rule and a frantic sacrifice. It collapses immediately, and predictably, because it was never rooted in anything the people had lived through. Transposed from knowing to changing, this is again fox against hedgehog: durable change is adaptive and earned, built from values already proven in experience, while hysteric change is rigid and unearned, a rule imposed and worshipped before anyone has reason to believe in it.

Chapters 4 and 5 belong to the pattern of leading and organizing. Take the king that the Little Prince visits: he commands a planet with nothing on it, and every one of his orders, on inspection, is an order he had already privately made sure would be obeyed before he ever gave it. His authority looks absolute and is actually hollow, because it has never once been tested against the real possibility of failing. Xenophon's Ten Thousand are the opposite case, and a harder one: a leaderless army of hired strangers, stripped of its generals by treachery, governs itself successfully – not because self-organization automatically beats command, but because by the time it had to rely on itself it had already earned its own cohesion the hard way, through Cunaxa, through the march, through the kind of shared ordeal Pirandello's frozen, mask-wearing characters never get to live through, and that Belyaev's foxes and Tolstoy's Nekhlyudov show, each in their own register, is the only thing that actually reshapes a disposition. It comes down again to earned authority against proclaimed authority, fox against hedgehog: the king's rule is unearned and brittle the moment it is tested, while the Ten Thousand's self-government is earned and survives the very catastrophe – the murder of its own commanders – that by every rule of hierarchy should have destroyed it.

7.2 What this actually asks of a working engineer or manager

It would be a mistake to leave the gallery above as the conclusion, because six restatements of fox versus hedgehog are not, by themselves, something a working software engineer or manager can do anything with on Monday morning. So let me compress the six chapters into the three patterns and say plainly what each one asks of someone actually running or working inside a software team.

Knowing, in practice, means epistemic humility as a working method, not a mood. It means answering «I don't know» or «I can't predict that» when that is the true answer, and then doing what Hippocrates did instead of what Galen did: reading the actual signs in front of you, case by case, revising your judgment as evidence comes in, and not letting a single adopted doctrine – Scrum, waterfall, whatever it calls itself – become immune by construction to what your own project is telling you.

Changing, in practice, means refusing the shortcut of the Golden Calf. It means values before principles, principles before practices, and it means telling people why a change matters in terms anchored in experience they already recognize as their own, instead of handing down a new rule and expecting compliance to follow from authority alone. A practice adopted without the values underneath it is a rule waiting to be melted down into an idol the moment the person who imposed it looks away.

Leading, in practice, means the opposite of what the king on his empty planet does: earning authority by giving direction that could actually fail, and visibly bearing the cost when it does, rather than hiding behind decisions quietly ratified in advance that can therefore never be shown to have been wrong. It also means resisting the opposite shortcut, the one that says self-organization simply wins by default. A team earns the right to govern itself only once it has actually built the cohesion self-organization presupposes, the way the Ten Thousand did, not because a manual or a manifesto says self-organizing teams are best.

Put all three together, in the fewest words I can manage: be a fox, not a hedgehog: manage uncertainty by adapting from values and lived experience, not by clinging to constructs that only give the illusion of predictability and control.

I do not mean this as a slogan, the kind of thing printed on a poster and then never thought about again. It is closer to a discipline, in the old sense in which Hippocrates' own art was a discipline: something practiced case by case, against a real temptation, in every sprint, every planning meeting, every difficult conversation with a stakeholder who wants a Gantt chart, to reach instead for the hedgehog's one big thing, because the one big thing is much more comfortable to hold onto than the fox's admission that the terrain has to be sensed anew every time. Nothing in this book makes that discipline easier to practice. If it does anything, it makes it easier to see the hedgehog coming, whatever costume it is wearing this time, whether it calls itself a methodology, a maturity model, or simply common sense.

7.3 The roots

I called this book *At the roots of lean and agility*, and I want to close by taking that title literally. Six chapters and this short seventh one keep going back, past the vocabulary of sprints and backlogs and burndown charts, to sources that have nothing to do with software and everything to do with what it means to know, to change, and to lead. Socrates, standing trial for confessing his own ignorance. The Exodus narrative, with its patient order of values before principles before practices, and its cautionary idol melted from borrowed gold. Cicero, writing home to a sick friend with a remedy grounded in nothing more than lived experience, centuries before anyone could explain why it worked. Xenophon's Ten Thousand, electing their own generals in the middle of hostile territory because no one else was left to appoint them. Hippocrates, reading a fever hour by hour instead of reaching for a system with an answer for everything. And closer to our own time: Belyaev's foxes turning tame within a handful of generations; Tolstoy's Nekhlyudov, changed not by decree but by what he could no longer refuse to see; Pirandello's six characters, arriving on stage already wearing masks no director's plan can script away; Saint-Exupéry's small, hollow king, ruling absolutely over nothing at all. None of these voices knew anything about software. That is exactly the point. They are not analogies borrowed to decorate an argument I could have made some other way – they are the roots themselves, the old and various places where the human understanding of uncertainty, of durable change, and of earned authority was worked out long before anyone had a computer to run it on. Lean and agile software engineering did not invent this understanding. It rediscovered it and dressed it, for our own century, in the vocabulary of planning games, retrospectives, self-organizing teams. If this book has done what I hoped, the reader who finishes it will no longer hear that vocabulary as a confession of insufficient rigor, but as what it has actually been all along: a late, faithful branch of a very old root.



Figure 7.3: The fox and the hedgehog, the ship and the pillars, the idol and the throne, gathered in one landscape.

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